(Volga or Kazan) Tatar

Introduction

Tatar (ISO 639-3 `tat`, Glottolog `tata1255`; also Kazan Tatar, Volga Tatar) is a Kipchak or (North-)Western[1] Turkic language. The majority of Tatar speakers are located in the Russian Federation (3.26 million at the time of the 2021 Census), largely in the federal Republic of Tatarstan, where Tatar has official status; the variety of Kazan, the capital of Tatarstan, is the principal basis for the literary standard. Tatar is also spoken in the neighbouring republics (Bashkortostan, Mordvinia, Mari El, Chuvashia), altogether a large area stretching from the Volga past the Urals, and Tatar speakers are the largest linguistic minority within the Russian Federation. Significant Tatar-speaking populations can also be found through much of Central Asia and the former Soviet Union (Kazakhstan, Uzbekistan, Ukraine, Turkmenistan, Kyrgyzstan).

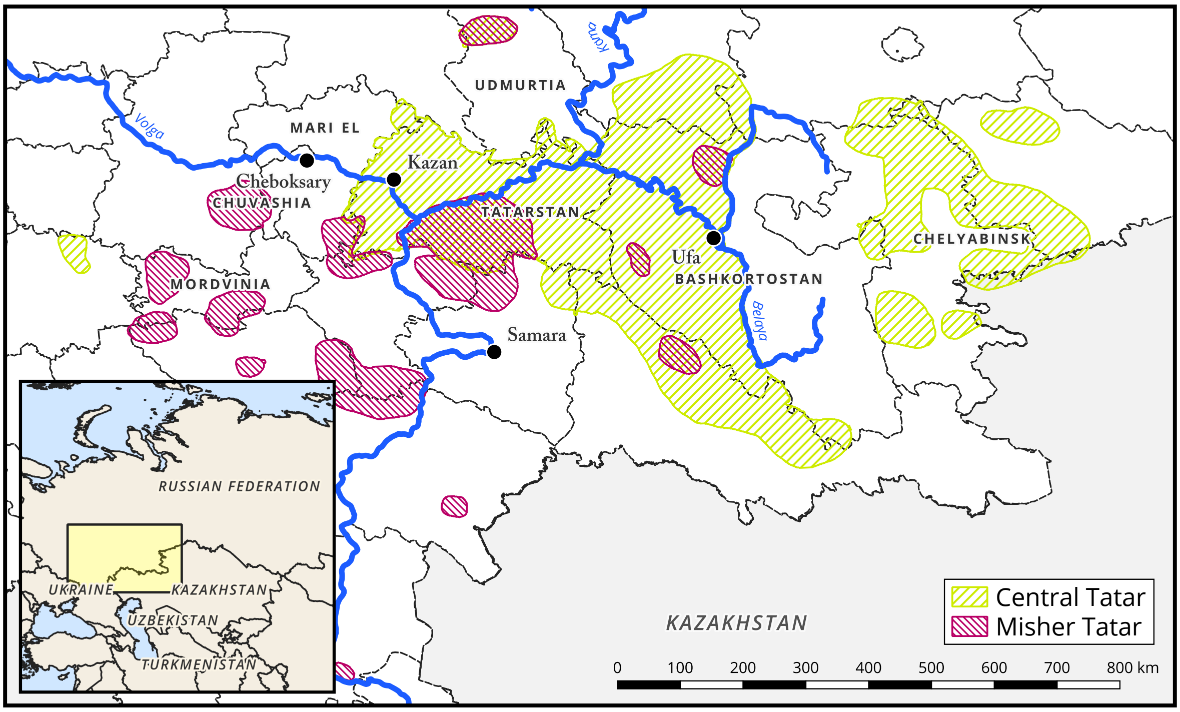


Figure 1: Major dialect groups within the Tatar-speaking area, based on Makhmutov (1969).

It is challenging to estimate the total number of L1 speakers of Tatar outside the Russian Federation, and thus the total number of speakers of the language overall. The total population of the ethnic group is estimated at 5.3 to 6 million, of which 2 million reside in Tatarstan itself and about 4 million in the Russian Federation as a whole (Károly 2022: 178); but the number of

[1] Most classifications of the Turkic languages use geographical directions as identifiers for the branches of the family. Accordingly, the Kipchak group is West Turkic for Benzing (1959: 2), the Northwest-division for Menges (1959: 6), and the Northwestern branch for Johanson (2021: 22). For Károly (2022: 12–18), the Siberian Turkic languages (excepting Sakha and Dolgan) are 'Central', and Kipchak 'Western', with Tatar belonging to a 'Northern', or 'Volga-Ural' sub-branch of the latter.

speakers of the language is likely to be somewhat lower than this, due to ongoing and increasing shift to Russian among Tatars inside the Russian Federation.

Tatar is traditionally divided into two dialect groups, central and western; see Berta (1989: 42–50) for dialectological details. The western group, jointly referred to as Misher (occasionally Mishar) Tatar, is fairly heterogeneous (Eleusin 2022: 10–13), and is not treated here; speakers of these varieties do not typically identify their variety with the central varieties. The central group includes the Kazan variety, on which the formal standard is largely based. 'Siberian Tatar', often considered to be an eastern Tatar group, is in fact a group of West Siberian Turkic varieties (Tobol-Irtysh, Baraba, and Tom); these have converged to Tatar itself to various degrees during the twentieth century, but are rather distant from a phylogenetic point of view. Tatar is very closely related to the neighbouring Bashkir, and is fairly close (although less so) to Kazakh, which is perhaps the Kipchak language best known to phoneticians and phonologists.

General descriptive overviews of the language, largely without supporting acoustic data, are given by Poppe (1963, 1968); Faseev (1966); Kurbatov et al. (1969); Comrie (1997); Berta (1998); Şahan Güney (2015); Burbiel (2018); Berta (2022). Early instrumental descriptions are given by Bogoroditskiĭ (1934: 27–79; 1953: 38–116), and by Baĭchura (1960); the descriptive grammar of Zakiev (1993: 54–87) includes a chapter on acoustic and articulatory phonetics. Attention to Tatar in the phonetic (and phonological) literature has been limited, but up-to-date work in this respect includes the descriptions of the vowel system and harmony system(s) appearing in Conklin (2015a,b, 2019); Conklin & Dmitrieva (2018, 2019), and the characterisation of the prosodic and intonational systems in Royer (2017); Royer & Jun (2018, 2019).

Data

This illustration represents primary audio material recorded from 9[2] speakers of Tatar (2 male, 7 female), years of birth 1964–2005 (median 1989, SD 16); more detailed metadata, including geographical origin and non-Tatar language competence, appear in Table 1. All speakers reported using Tatar daily and considered it to be their primary language across activities, including the one speaker (F07) resident outside Tatarstan.

All speakers read a list of 211 words in isolation. 5 speakers (F03, F04, F05, F06, and M02) read a further 115 words in isolation, 44 words in the carrier phrase ⟨Ул [X] сүзен кабатлады⟩ /ul [X] sʉzen qɑbɑt̠lɑd̠ə/[3] 'S/he repeated the word [X]' in order to characterise word-level stress accent, a short dialogue to elicit sentence-level intonation, and the folk tale ⟨Куркак юлдаш⟩ 'The cowardly companion'[4]. Our main consultant, F07, read 57 further words in isolation, a further 26 words in the carrier sentence, 20 further sentences for more comprehensive

[2]A supplementary tenth recording, of a speaker F00 (born 1949 or 1950) reading a short passage was also referenced for confirmation; this speaker did not produce the wordlist(s), and so is not included in summary statistics. The passage referenced, together with the corresponding audio, is available in full in Károly (2022).

[3]Adapted from the experiment in McCollum & Chen (2020).

[4]This traditional story, often used in Tatar schools (Zakirova 2011: 145), was chosen as a culturally-grounded alternative to the more widely-used 'The North Wind and the Sun' passage.

Table 1: Speaker metadata (ID, year of birth, place of origin, language competence).

Speaker ID	Year of birth	Year of recording	Place of origin	Other languages
M01	1997	2022	Sabinskiĭ district	RU, EN, DE, TR, AR
F01	2003	2022	Sabinskiĭ district	RU, EN, DE
F02	1979	2024	Tyulyachi → Kazan	RU, EN, BA
F03	1991	2022 + 2024	Naberezhnȳe Chelnȳ → Kazan	RU, DE
F04	1988	2024	Kazan	RU
F05	2004	2024	Rȳbno-Slobodskiĭ district → Kazan	RU, EN
F06	1966	2024	Kukmor → Kazan	RU
M02	2005	2024	Kazan	RU, EN
F07	1964	2025	Verkhniĭ Baĭlar / Naberezhnȳe Chelnȳ → Kazan (→ Stockholm, Sweden)	RU, SV, ES, EN, TR
(F00	1949/1950	2016	Nizhnee Gareevo → Kazan	RU)

coverage of patterns of sentence-level intonation, and a second folk tale, ⟨Күбәләкләр ничек барлыкка килгән?⟩ ‘How butterflies came to be’. Final token numbers may vary due to repetitions, variant pronunciations, hesitations and speech errors, and occasionally due to background noise.

Data collection and quality

Data collection was subject to severe limitations, due to ongoing and indefinite restrictions on travel to the region by foreign researchers. Speakers M01, F01, F02 were recorded remotely in Tatarstan with unknown equipment, presumably smartphones, by the speakers themselves, and evaluated for quality by the authors. Speakers F03, F04, F05, F06, and M02 were recorded in Kazan with a Zoom H1N recorder (16 bits, 44.1 kHz) and on-board microphone. Speaker F07 was recorded in a quiet environment in Stockholm, Sweden, with a Zoom H5 recorder (16 bit, 44.1 kHz) and a Røde Lavalier GO lapel microphone. Speakers F03, F04, F05, F06, M02, and F07 gave short extemporaneous introductions in Tatar before reading, which partially mitigates the effects of experimenter languages (Russian, Swedish); these do not constitute part of the dataset. Recordings were evaluated for quality by marking silences and calculating the signal-to-noise ratio (SNR), $20\log_{10}\frac{\text{signal max amplitude}}{\text{noise max amplitude}}$ as a measure of background-noise-driven inaccuracy (see e.g. Sanker et al. 2021; Nance et al. 2024); four passed the conservative 30 dB SNR threshold suggested as the clinical gold standard by Deliyski et al. (2005) (M01 34.08 dB; F01 33.00 dB; F03 34.77 dB; F07 39.52 dB), and the remainder passed the 20 dB mark (F02 23.27 dB; F04 28.30 dB; F05 28.38 dB; F06 28.06 dB; M02 23.47 dB), where 10 dB SNR is a reasonable cut-off point for unacceptably noisy recording for sociophonetic purposes per De Decker (2016). SNRs were artificially depressed by breaths, since these were not distinguished from true silences in tagging. As such, we consider these recordings suitable for analysis.

Summary statistics cover all recordings, but due to the conditions of data collection, the audio for F07 is the highest-quality and best-controlled, and should be treated as our primary reference data. All audio files provided with this paper are labelled with the speaker code.

Consonants

Consonantal inventory

We describe 26 distinct consonantal phonemes for Tatar. Sounds to which cautions regarding allophonic status should apply have been marked (∗). Those with marginal status, found exclusively in Arabic, Persian, or Russian loanwords, have been marked (‡); these were variably produced.

	Bilabial	Labiodental	Dental	Post-alveolar	Palatal	Velar	Uvular	Glottal
Plosive	p b		t̪ d̪			k g	q*	
Fricative		f v‡	s z	ʃ ʒ	ɕ ʑ		χ ʁ*	h‡
Affricate			t͡s‡					
Nasal	m		n				ŋ	
Trill				r				
Approximant	w				j			
Lateral appx.			l					

p	#_	[pə.ˈɕ̻ɑq]	⟨пычак⟩	‘knife’
	V_V	[ɒ.ˈpɒ]	⟨апа⟩	‘aunt’
b	#_	[bu]	⟨бу⟩	‘this’
	V_V	[ɒ.ˈbəj]	⟨абый⟩	‘older brother’
	_#	[kɫub̥]	⟨клуб⟩	‘club’
t̪	#_	[t̪ʰɑʑ̥]	⟨таж⟩	‘crown’
	V_V	[ɒ.ˈt̪ɒ]	⟨ата⟩	‘father’
	_#	[ɒt̪]	⟨ат⟩	‘horse; name’
d̪	#_	[d̪ɒ.ˈɫɒ]	⟨дала⟩	‘flat land’
	V_V	[ɒ.ˈd̪əm]	⟨адым⟩	‘step’
	_#	[ʃə.kə.ˈɫɑd̪̥]	⟨шоколад⟩	‘chocolate’
k	#_	[kø̞z̪̥]	⟨көз⟩	‘autumn’
	V_V	[i.ˈke]	⟨ике⟩	‘two’
	_#	[kʉk]	⟨күк⟩	‘sky’
g	#_	[gʉl]	⟨гөл⟩	‘flower’
	V_V	[si.ˈgez̪̥]	⟨сигез⟩	‘eight’
	_#	[pi.ˈrog̊]	⟨пирог⟩	‘pirog’
q	#_	[qɒz̪̥]	⟨каз⟩	‘goose’
	V_V	[ɒ.ˈqəɫ]	⟨акыл⟩	‘reason’
	_#	[qu.ˈnɑq]	⟨кунак⟩	‘guest’
f	#_	[fæn]	⟨фəн⟩	‘science’
	V_V	[læ.ˈfəz̪̥]	⟨лəфыз⟩	‘word’
	_#	[χæ.ˈref]	⟨хəреф⟩	‘letter’
v	#_	[vəg.ˈzɑɫ]	⟨вокзал⟩	‘train station’
	V_V	[zɑ.ˈvod̪̥]	⟨завод⟩	‘factory’
	_#	[kə.lʲek.ˈt̪iv̥]	⟨коллектив⟩	‘team’
s	#_	[sɒp]	⟨сап⟩	‘handle’
	V_V	[ɒ.ˈsɒt̪]	⟨асат⟩	‘easy’
	_#	[d̪us]	⟨дус⟩	‘friend’
z	#_	[(ɒ̆)ˈzɒt̪]	⟨зат⟩	‘essence’
	V_V	[t̪ɒ.ˈzɒ]	⟨таза⟩	‘clean’
	_#	[ɒz]	⟨аз⟩	‘few’
ʃ	#_	[ʃuɫ]	⟨шул⟩	‘that’
	V_V	[ɒ.ˈʃɒw]	⟨ашау⟩	‘eat-VN’
	_#	[t̪əʃ]	⟨тыш⟩	‘outside’
ʒ	#_	[ʒɑˈnər]	⟨жанр⟩	‘genre’
	V_V	[ɒ.ˈʒur]	⟨ажур⟩	‘openwork’
	_#	[st̪ɑʒ]	⟨стаж⟩	‘experience’
ɕ	#_	[ɕ̻ɒq]	⟨чак⟩	‘time’
	V_V	[ɒ.ˈɕ̻ə]	⟨ачы⟩	‘bitter’
	_#	[ɒɕ̻]	⟨ач⟩	‘hungry’
ʑ	#_	[ʑ̻ir]	⟨җир⟩	‘place’
	V_V	[ə.ri.ˈʑ̻ɑ]	⟨риҗа⟩	‘hope’
	_#	[t̪ɒʑ̻]	⟨таҗ⟩	‘crown’
χ	#_	[χɒt̪]	⟨хат⟩	‘line’
	V_V	[ɒ.ˈχər]	⟨ахыр⟩	‘end’
	_#	[ˈruχ]	⟨рух⟩	‘soul’

ʁ	#_	[(ŏ)ʁɒ.ˈsər]	⟨гасыр⟩	‘century’	ŋ	V_V	[ɒ.ˈŋɒ]	⟨аңа⟩	‘3s.DAT’
	V_V	[ɒ.ˈʁɒɕ]	⟨агач⟩	‘tree’		_#	[ɒŋ]	⟨аң⟩	‘reason’
h	#_	[hær̥]	⟨һәр⟩	‘each’	w	#_	[wæ]	⟨вә⟩	‘and’
	V_V	[ʃæ.ˈhær]	⟨шәһәр⟩	‘city’		V_V	[ɒ.ˈwəz]	⟨авыз⟩	‘mouth’
	_#	[gø.ˈnɑh]	⟨гөнаһ⟩	‘sin’		_#	[sɒw]	⟨сау⟩	‘healthy’
t͡s	#_	[ˈt͡selʲ]	⟨цель⟩	‘goal’	j	#_	[jɒz]	⟨яз⟩	‘spring’
	V_V	[li.ˈt͡sej]	⟨лицей⟩	‘high school’		V_V	[ɒ.ˈjɒq]	⟨аяк⟩	‘foot’
	_#	[ne.ˈmet͡s]	⟨немец⟩	‘German’		_#	[bɒj]	⟨бай⟩	‘wealthy’
m	#_	[mif]	⟨миф⟩	‘myth’	l	#_	[ɫɒj]	⟨лай⟩	‘dirt’
	V_V	[ɒ.ˈmir]	⟨амир⟩	‘commander’		V_V	[ɒ.ˈɫɒ]	⟨ала⟩	‘striped’
	_#	[t̪æm]	⟨тәм⟩	‘taste’		_#	[bɒɫ]	⟨бал⟩	‘honey’
n	#_	[ni]	⟨ни⟩	‘what’	r	#_	[rus]	⟨рус⟩	‘Russian’
	V_V	[ɒ.ˈnɒ]	⟨ана⟩	‘mother’		V_V	[bɒ.ˈru]	⟨бару⟩	‘go-VN’
	_#	[d̪in]	⟨дин⟩	‘religion’		_#	[ber]	⟨бер⟩	‘one’

Previous descriptive work has suggested between 24 and 30 phonemic consonants, with major points of difference being the identification of allophonic variation, and the inclusion of marginal segments found exclusively in loanwords (Burbiel 2018: 5–8; Faseev 1966: 816–817; Kurbatov et al. 1969: 88–96; Zakiev 1993: 60–63, 91–96; Comrie 1997: 901–902).

In the first instance, in Tatar, as is the case in essentially all the Turkic languages, much of the consonantal inventory is subject to significant conditioned variation, dependent primarily on the backness (and roundness) of adjacent vowels (Zakiev 1993: 94–95)[5] but also on position within the syllable. As such, it can be difficult to distinguish between alternations involving multiple phonemes, and those involving allophones of a single phonemic consonant. This analytic difficulty is reflected in the literature; Burbiel (2018: 5–8) and Comrie (1997: 901–902) give four contrastive phonemes /q/ /ɣ/ /k/ /g/, Poppe (1968: 10–13) and Baskakov (1988: 137) give [q] and [ɣ] as allophones of [k] and [g] respectively, and Şahan Güney (2015: 13–19) gives phonemic /ʁ/ and /g/, but allophonic [q] /k/. We treat the situation of the velars and uvulars as phonemic rather than allophonic, due to the existence of a small number of unpredictably-distributed sounds (details are given in the discussion of plosives below); however, we take most other front–back patterns to be allophonic (due to the lack of such unpredictability).

In the second instance, the literature highlights /v/ ⟨в⟩ and /t͡s/ ⟨ц⟩, which appear only in Russian loans; /h/, which appears only in borrowings from Persian and Arabic; and a purported /ʔ/, described as only being attested in words of Arabic origin. Most of our speakers produced [v] in transparent Russian borrowings e.g. ⟨вокзал⟩ [vəg.zɑɫ] ‘train station’ (although not all; speakers F01 and F07 produced [b] in all such Russian borrowings), but all speakers had [w] in Arabic and Persian borrowings e.g. ⟨вәгъдә⟩ [wæʁ.ˈd̪æ] ‘promise’, ⟨һава⟩ [hɒ.ˈwɒ] ‘air’. [h] is variable both between speakers and within individual speakers; F03 has ⟨һәр⟩ [h] [hær] ‘each’

[5] Zakiev 1993: 94–95 attributes this to Russian influence, but similar patterns are found across the family, including in languages such as Turkish which have had next to no contact with Russian.

but ⟨аждаһа⟩ [χ] [ɒʒ.d̪ɒ.ˈχɑ] ‘dragon’, and F06 has ⟨гөнаһ⟩ [χ] [gø̞.ˈnɑχ] ‘sin’, where speaker F07 had [h] for all of these items. ⟨ц⟩ [t͡s] appears only in Russian loans, and not all speakers produce it: ⟨цирк⟩ [s] [sirk] [t͡s] [t͡sirk] (F01) ‘circus’, ⟨немец⟩ [t͡s] [ˈne.met͡s]~[ˈnʲe.met͡s], but for F05 [ˈnʲe.mes] ‘German’. ⟨ф⟩ [f] and ⟨ж⟩ [ʒ] are also restricted to borrowings, but unlike the consonants we mark as marginal, are produced by all speakers and in all words without exception.

We have decided against including a glottal plosive /ʔ/ in the final inventory. The glottal plosive appears only where vowel hiatus is avoided, and was not produced by all speakers, as illustrated in Figure 2 for ⟨тәэсир⟩ [t̪æʔsir]~[t̪æesir] ‘influence’. Some speakers also produce a clear glottal stop before word-initial vowels, including epenthetic vowels, but this has no contrastive status either.

A proposed long /ɕː/ from Russian (Burbiel 2018: 7; Faseev 1966: 816; Zakiev 1993: 92; Şahan Güney 2015: 14) is not consistently contrastively distinguished from /ɕ/ by duration: the initial consonant in ⟨щётка⟩ [ˈɕ(ː)ɵt̪.kɑ] ‘brush’ was produced with a mean duration of 180ms across all speakers (SD 44.26, $N = 9$), versus a mean of 162ms (SD 39.33) for all tokens of word-(utterance-)initial /ɕ/ in isolation ($N = 36$, measured manually).

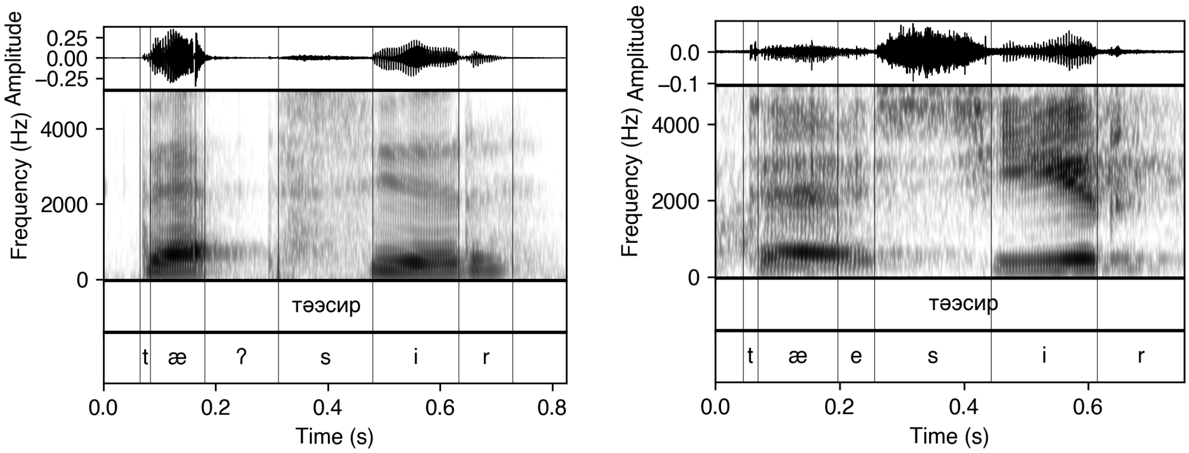


Figure 2: Waveforms and spectra for two speakers’ productions of ⟨тәэсир⟩ ‘influence’: F07 (left) with the glottal plosive, F05 (right) without.

Obstruents (plosives, fricatives, affricates)

Plosives

We describe seven plosives for Tatar, four voiceless /p t̪ k q/ and three voiced /b d̪ g/.

In the lexicon, there is an alternation between velar and uvular based on the backness of the surrounding vowels, as in the vast majority of the Turkic languages (see e.g. McCollum & Chen 2020: 278 for Kazakh): the typical pattern is that /k/, /g/ occur with front vowels, corresponding to /q/ (which can have a fricated or affricated release [q͡χ]) and typically fully-fricative /ʁ/, though rarely also the uvular plosive [ɢ][6], ⟨гакылның⟩ /ʁɑqəlnəŋ/ [ɢɒ.qəɫ.ˈnəŋ] 'mind-GEN' (F02), with the back vowels. Although a front-back alternation can be seen to be synchronically active in the morphophonology, as shown for ⟨балаларга⟩ [bɒ.ɫɒ.ɫɑr.ˈʁɑ] 'child-PL-DAT' and ⟨бүтәннәргә⟩ [bʉ.t̪æn.nær.ˈɡæ] 'other-PL-DAT' in Figure 3, we treat both /q/ and /ʁ/ as phonemic due to unpredictable violations of the distributional pattern in loanwords: in borrowings, /q/ and /ʁ/ may occur together with front vowels, as in ⟨тәкъдир⟩ [t̪æq(æ̆).ˈd̪ir] 'fate, destiny', ⟨тәкъдим⟩ /t̪æqd̪im/ [t̪æχ(æ̆).ˈd̪im] 'offer, suggestion', ⟨диккәт⟩ [d̪iq.ˈqæt̪] 'attention', ⟨кәләм⟩ [qæ.ˈlæm] 'carpet', ⟨кәгазь⟩ [qæ.ˈʁɑz] 'paper'; but /k/ and /ɡ/ may also occur adjacent to back vowels in (Russian) borrowings, as in ⟨вагон⟩ [vɒ.ˈɡon] 'wagon', ⟨грамота⟩ [ˈɡrɑ.mə.t̪ɑ] 'literacy', ⟨коллектив⟩ [kə.lʲek.t̪iv] 'team', ⟨колхозның⟩ [kəɫ.χoz.ˈnəŋ] 'kolkhoz-GEN', ⟨шоколад⟩ [ʃə.kə.ɫɑd̪̥] 'chocolate'. The orthography only rarely indicates this contrast, but /q/ and /ʁ/ in front environments are sometimes marked with ⟨ъ⟩ in Arabic-Persian borrowings.

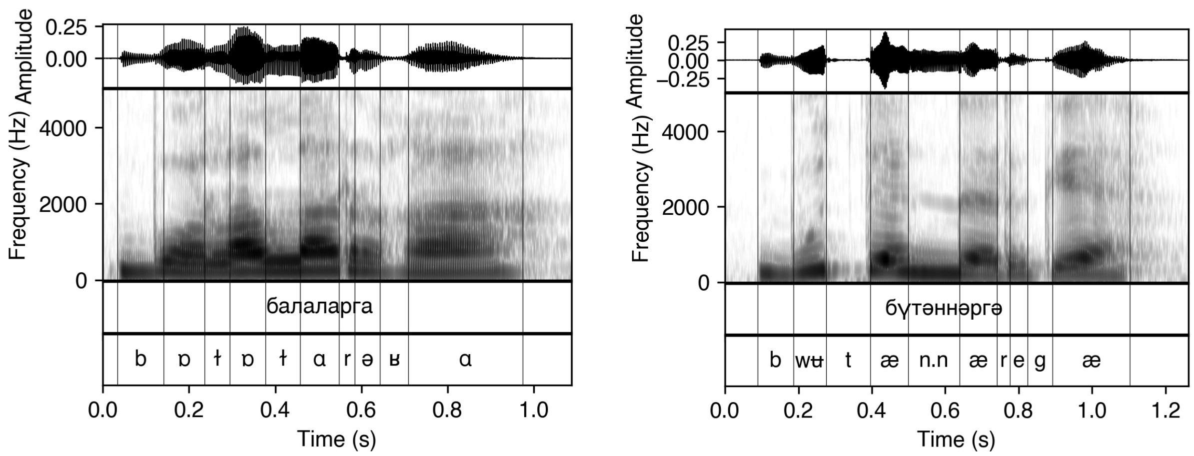


Figure 3: Waveforms and spectra for ⟨балаларга⟩ [bɒɫɒɫɑrəˈʁɑ] 'child-PL-DAT' and ⟨бүтәннәргә⟩ [bʷʉtæn:æɾeˈɡæ] 'other/stranger-PL-DAT' produced by speaker F07. The alternation in the dative suffix depends solely on vowel backness.

The mean voice onset time for the seven plosives in Tatar, by place of articulation, is shown in Table 2 below. The general tendency in Tatar is for voiceless plosives to be rather lightly aspirated (small +VOT)[7], and voiced plosives variably pre-voiced (-VOT). Most speakers produce a few tokens of the voiced series, especially /b/, with no pre-voicing and short positive VOT, but the majority of voiced tokens show a clear period of vocal fold vibration. Although

[6]Poppe (1968: 11) seems to suggest the voiced uvular plosive as a possibility, and it is certainly the case that the sound in question is historically a plosive. We choose the /ʁ/ as the representation in the inventory, however, as several of our speakers produce no [ɢ] at all, whereas there is no speaker that does not have [ʁ] in at least some environments. The voiced fricatives as a class are susceptible to epenthetic repairs in which /ʁ/ participates, which we discuss below.

[7]Claims made in the descriptive literature (Burbiel 2018: 6; Poppe 1968: 10) that the /t̪/ is only slightly aspirated, or less so than the other voiceless plosives, seem to derive solely from the well-established effect of place of articulation on VOT (Lisker & Abramson 1964). No language-specific effect beyond this is justified by the data.

the VOT contrast is naturally most detectable word-initially, few tokens of initial /g/ are available due to extensive spirantisation, both synchronic and diachronic (that is, /ʁ/ from historic /g/); we discuss 'weakening' and spirantisation in this position later in this section. Figure 4 shows all VOT values for word-initial plosives by speaker, and Table 2 shows mean VOT values across all tokens and speakers. /k/ and /q/ are slightly more post-aspirated than the other voiceless plosives; the voiced plosives show more across-the-board variability, both intra- and inter-speaker.

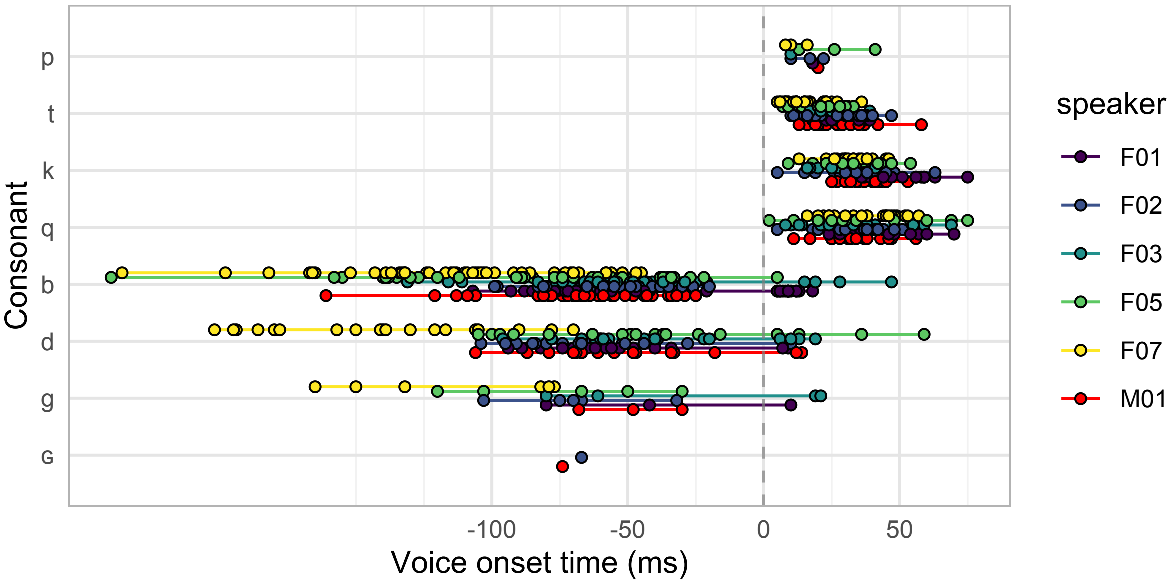


Figure 4: Voice-onset time (VOT) of 732 **word-initial** voiceless and voiced plosives, measured manually; VOT = 0 corresponds to the time of release. Plosive [ɢ] realisations of /ʁ/ are included for comparison.

Table 2: Voice onset time (VOT) for word-initial plosives by place of articulation, in ms.

	Voiceless plosives				Voiced plosives		
	Mean VOT (ms)	SD	Tokens		Mean VOT (ms)	SD	Tokens
p	+18	9	12	b	−67	44	245
t	+22	9	133	d	−66	51	106
k	+36	12	97	g	−68	46	26
q	+36	14	111	ɢ	−71	5	2
All	+29	13	353	*All*	−67	45	379

As shown in the table of consonants above, voiceless and voiced plosives contrast initially and, root-internally, in intervocalic position. Voiced plosives are not available in word-final (root-final) position natively, and appear to be subject to synchronic final fortition; utterance-final plosives, especially velars and uvulars, can have long, fricated releases. A small number of potentially surface-voiced word-final plosives, given in the table of consonant examples above, appear in recent Russian borrowings; these are subject to further complications due to the fact that Russian itself shows phonological final devoicing or final fortition (Yanushevskaya & Bunčić 2015; Kharlamov 2014). Kharlamov (2014) reports incomplete neutralisation of plosive voicing in Russian, with underlyingly voiced plosives partially distinguishable from underlyingly voiceless ones on the basis of both duration and presence of active voicing (i.e. glottal pulsing). We find variation both across and within speakers in Tatar, illustrated across all speakers in Figure 5: our reference speaker, F07, neutralises the contrast between /t̪/ and /d̪/ utterance-finally in ⟨шоколад⟩ [ʃəkəɫɑd̪̥] 'chocolate', but produces identifiable glottal pulsing in ⟨клуб⟩ [kɫub] 'club', although we note the possibility that incomplete neutralisation in our data is at least partially driven by the use of orthographic input (see, among many others, Mascaró 1987; Warner et al. 2006; Iverson & Salmons 2011).

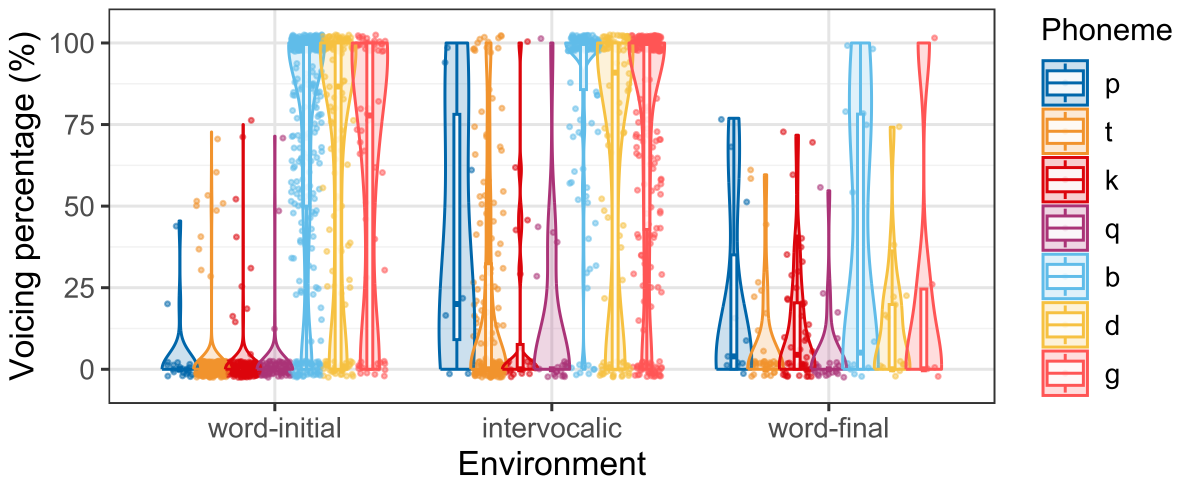


Figure 5: Voicing percentages, calculated automatically as percentage of voiced frames over the entire duration of the segment, for plosives (N = 1888) in words produced in isolation, in word/utterance-initial, intervocalic root-internal, and word/utterance-final positions.

The 'intervocalic' category in Figure 5 shows only root-internal plosives. In fact, root-final /p/ and /k/, but not /t̪/, voice (and possibly lenite, discussed below) intervocalically or when followed by an initial voiced stop across both morpheme and word boundaries, as illustrated in Figure 6; voicing at morpheme boundaries is represented consistently in the orthography. At the word level, compare ⟨бишегеңдә⟩ /biʃek-eŋ-d̪æ/ [bi.ʃe.geŋ.ˈd̪æ] (F06) [bi.ʃe.ɰeŋ.ˈd̪æ] (F07) *[bi.ʃe.keŋ.ˈd̪æ] 'cradle-POSS.2S-LOC', ⟨аулагында⟩ /ɑwlɑk-ən-d̪ɑ/ [ɒw.ɫɒ.ʁən.ˈd̪ɑ] *[ɒw.ɫɒ.kən.ˈd̪ɑ] 'courtyard-POSS.3S-LOC', but ⟨хөрмәтенә⟩ /χø̞rmæt̪-en-æ/ [χø̞r.mæ.t̪e.ˈnæ] *[χø̞r.mæ.d̪e.ˈnæ] (F06, F07) 'respect-POSS.3S-DAT'. At the phrase level, consider ⟨җавап биргәли⟩ [ʑ̊ɒwɒb‿birgæli] 'answer give-VADV'; ⟨китап⟩ /kit̪ɑb/ [kit̪ɑp] but ⟨китап укый⟩ [kit̪ɑb‿uqəj] ~ [kit̪ɑβ‿uqəj] 'book read-PRS.3S'.

This framing implies no true underlying voicing in these root-final stops. An alternate analysis of these facts is as an underlying voicing contrast neutralised finally and recovered in intervocalic position; we cannot distinguish between these hypotheses on the basis of this dataset, but we are not aware of any consistent lexical exceptions to the pattern above that would provide positive evidence for an analysis of the latter type.

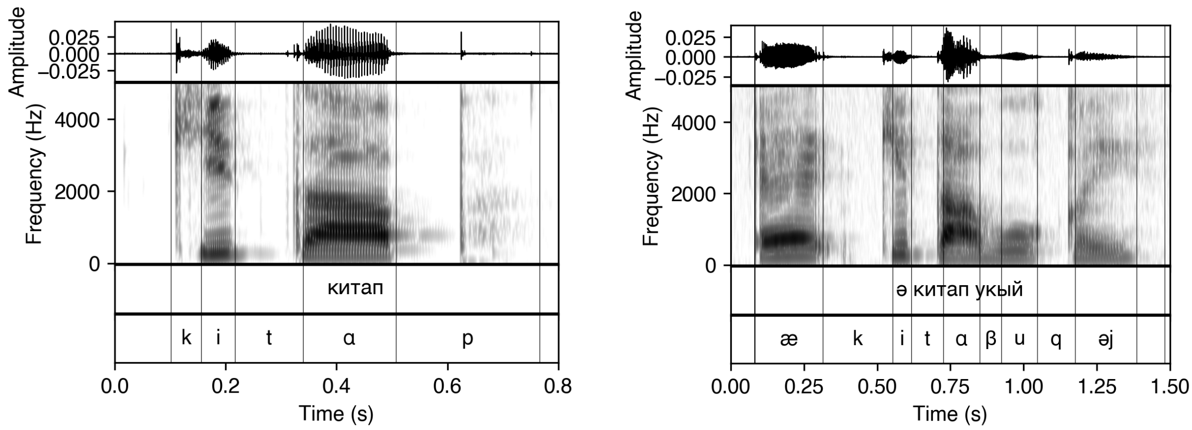


Figure 6: Waveforms and spectra for ⟨китап⟩ /kitɑb/ [kitɑp] and ⟨китап укый⟩ /kitɑb#uqəj/ [kitɑβ#uqəj] 'book read-PRS.3s', produced by speaker F07.

Word-initial and intervocalic plosives are often and highly variably lenited, typically spirantised[8] although occasionally lenited further to approximants; the application of this is dependent on underlying voicing, position, place of articulation, and vowel quality. Lenition is least frequent with the voiceless non-back plosives: we find no examples of fully fricative /t̪/ or /p/, but sporadic spirantisation of /d̪/ to [ð], as in ⟨гыйбадәт⟩ [ə.ʁəj.bɑ.ˈðæt̪] 'worship'. /b/ does not often spirantise domain-initially, although the burst can be notably weak (as in F05's ⟨бар⟩ [bɒr] 'existing'), but it may spirantise if intervocalic across a word boundary, ⟨Яхшы ау булсын!⟩ (F07) [jɒχ.ˈʃə ɒw‿βuɫsən] 'good hunt become-VOL.3s', or root-internally, ⟨абыйсы⟩ (F05) /ɑbəj-sə/ [ɒβəjsə] 'older.brother-POSS.3s' (and see Figure 6). /k/ and /q/ may spirantise word-initially or intervocalically; Figure 7 illustrates word-medial spirantisation of /q/ in ⟨тәкъдим⟩ /t̪æqd̪im/ [t̪æ.χæ̆.ˈd̪im] 'offer, suggestion' and ⟨бармакларында⟩ /bɑrmɑq-ɫɑr-ən-d̪ɑ/ [bɒrmɒχɫɑrənd̪ɑ] 'finger-PL-POSS.3S-LOC' produced by speakers F07 and F04 respectively. /g/ is rarely found in back vowel environments; in front vowel environments, it often spirantises to a velar /ɣ/, reliably distinguished from /ʁ/ by centre of gravity ($t = -5.91$, $p \ll .0001$, Cohen's $d = 0.608$; [ɣ] mean = 2464 Hz, SD = 988, $n = 155$; /ʁ/ mean = 1852 Hz, SD = 975, $n = 235$. These measurements are discussed further below for the true fricatives).

To quantify this, two acoustic measurements of lenition were considered. Harmonics-to-noise ratio (HNR), previously used by Broś et al. (2021), represents the degree of periodicity of a sound as the ratio of periodic to aperiodic energy in decibels; higher values of HNR reflect greater periodicity, and thus both voicing and burst weakness or absence. For each consonantal segment, HNR was calculated using Praat's harmonicity algorithm (implemented via Python

[8]It is presumably the phonologisation of this effect that has given rise to the phonemic /ʁ/.

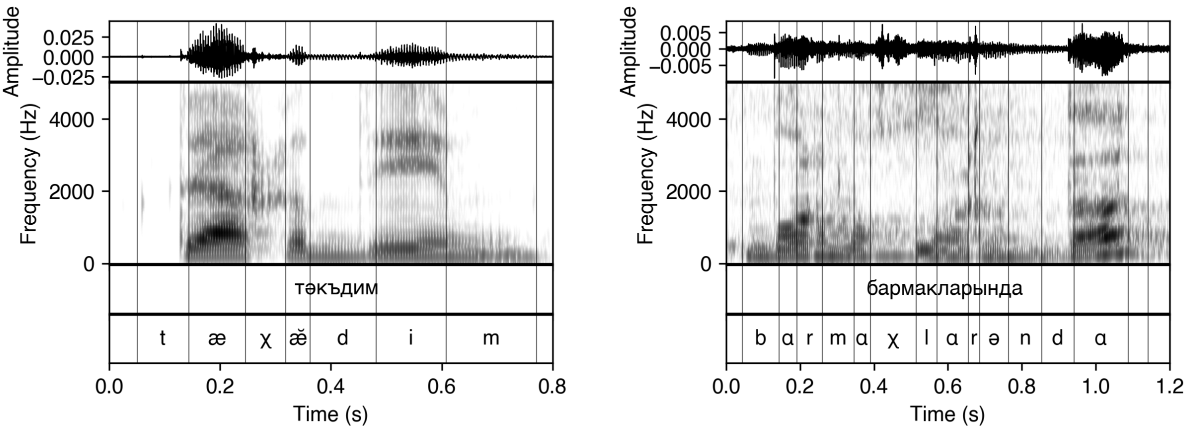


Figure 7: Waveforms and spectra for ⟨тәкъдим⟩ [t̪æ.χæ̆.ˈd̪im] 'offer, suggestion', produced by speaker F07, and ⟨бармакларында⟩ [bɒrmɒχɫɑrənd̪ɑ] 'finger-PL-POSS.3S-LOC', produced by speaker F04.

parselmouth) across the entire segmental duration using a 10 ms analysis window, with a minimum pitch of 75 Hz and silence threshold of 0.09. Intensity difference (established for lenition variously by Hualde et al. 2010; Parrell 2010; Martínez-Celdrán & Regueira 2008; Figueroa Candia & Evans 2015; Broś et al. 2021, among others) was calculated by extracting the minimum intensity (RMS dB) during the consonantal segment from the preceding segment's maximum intensity for final and intervocalic tokens, and for the following segment, in the case of word-initial tokens; lower values reflect higher sonority, or the reduction in acoustic prominence that accompanies weakened constriction. The results of both measures are visualised in Figure 8: intervocalic plosives, across the board, show increased harmonics-to-noise ratio and reduced intensity difference (that is, higher intensity) compared to word-initial plosives. (Since intervals containing word-final plosives often mark solely a long, fricated burst, it is not surprising that the harmonics-to-noise ratio and intensity difference measures disagree slightly here.) Both measures convolve the effects of voicing and the effects of spirantisation to some degree; with reference to Figure 5, however, we may attribute a considerable proportion of the effect in voiceless plosives to spirantisation, since the majority of underlying voiceless plosives do retain their voicelessness intervocalically.

Statistical analysis was conducted using mixed-effects linear regression models fitted with the `lme4` package (Bates et al. 2015) in R, with position (word-initial pre-vocalic vs. intervocalic, contrast-coded as −0.5 and 0.5 respectively), underlying voicing category (voiceless vs. voiced), and their interaction as fixed effects, and random intercepts for speaker and phoneme. p-values were obtained using the Satterthwaite approximation implemented in `lmerTest` (Kuznetsova et al. 2017). For HNR there was a significant main effect of position ($\beta = 1.999$, SE = 0.510, $t(1508) = 3.921$, $p < .001$) and a significant interaction with voicing ($\beta = -1.354$, SE = 0.607, $t(1415) = -2.232$, $p = 0.026$), with a larger effect for voiceless plosives ($\beta = 1.999$; voiced $\beta = 0.645$). For intensity difference, there was likewise a significant main effect of position ($\beta = -5.191$, SE = 0.674, $t(1504) = -7.707$, $p < .001$), though not of underlying voicing; both voiceless ($\beta = -5.191$) and voiced plosives ($\beta = -4.736$) showed substantial reductions in intensity difference intervocalically.

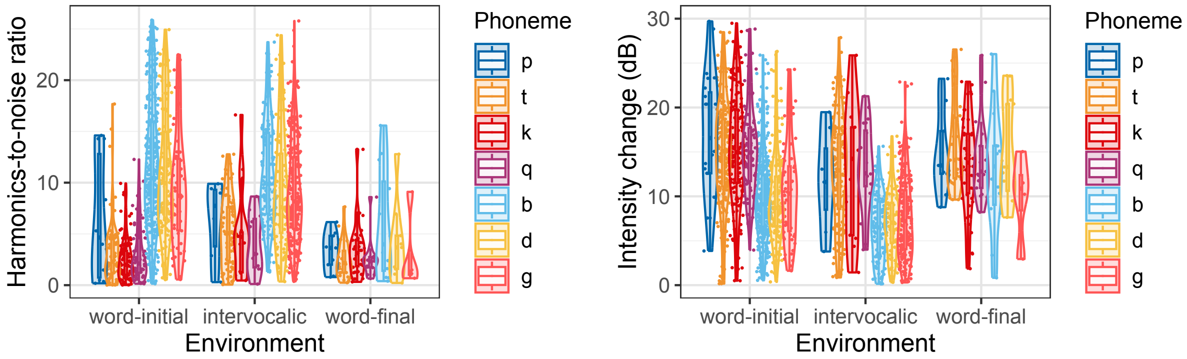


Figure 8: Harmonics-to-noise ratio (left) and intensity change (right) for underlying plosives (N = 1888) in words produced in isolation, in word/utterance-initial, intervocalic, and word/utterance-final positions.

As in Kazakh (McCollum & Chen 2020), affix-initial plosives in Tatar nasalise if and only if flanked by two nasals: that is, if preceded by a root-final nasal and followed by a nasal across the suffix vowel; compare the locative /-d̪(ɑ/æ)/, ⟨базарларда⟩ /bɑzɑr-lɑr-d̪ɑ/ [bɒ.zɒr.ɫɑr.ˈd̪ɑ] ‘market-PL-LOC’, ⟨телендә⟩ /t̪el-en-d̪æ/ [t̪e.len.ˈd̪æ] ‘language-POSS.3s-LOC’, and the ablative /-d̪(ɑ/æ)n/, ⟨өченчедән⟩ /ø̞ɕenɕe-d̪æn/ [ø̞.ɕø̞n.ɕø̞.ˈd̪æn] ‘third-ABL’, ⟨башкалардан⟩ /bɑʃqɑ-lɑr-d̪ɑn/ [bɒʃ.qɒ.ɫɑr.ˈd̪ɑn] ‘other-PL-ABL’, but ⟨изүемнән⟩ /izʉ-em-d̪æn/ [i.zʉ.em.ˈnæn] ‘chest-POSS.1s-ABL’. In Kazakh, this alternation applies to a wider range of affixes due to neutralisation of the plosive-nasal contrast in post-consonantal onsets; this does not occur in Tatar, in which the nasal-initial genitive /-n(ə/e)n/ is always nasal-initial on the surface, irrespective of the root-final consonant.

Fricatives and affricates

Tatar distinguishes six voiceless fricatives, /f s ʃ ɕ χ h/, and five voiced ones, /v z ʒ ʑ ʁ/. Voiced and voiceless fricatives contrast in all places of articulation except glottal.

We find evidence for only one affricate, /t͡s/, found only in recent Russian borrowings and variably realised as either the affricate or the stop. Historic /tʃ/ and /dʒ/, as found in related languages, are reflected in the orthography[9] (⟨ч⟩ /ɕ/ < /t͡ʃ/ and ⟨ж⟩ /ʑ/ < /d͡ʒ/; compare ⟨ш⟩ /ʃ/, ⟨ж⟩ /ʒ/) but have deaffricated to /ɕ/ and /ʑ/ in the modern language. We describe these sounds as alveolo-palatal, as does Comrie (1997: 902)[10]. There are a few cases in which surface

[9]Using graphemes that in other Cyrillic-orthography languages do correspond to affricates; we suspect that this gives rise to some erroneous reports of affricates in the present-day language.

[10]Zakiev (1993: 63) takes the position that these are non-sibilant fricatives (literally ‘однофокусный’ vs. ‘двухфокусный’), but agrees that they are not affricates, as do Kurbatov et al. (1969: 89) and Faseev (1966: 816). Poppe (1968: 12–13) does refer to them as affricates, but describes ‘weak’ stop elements due to incomplete closure (as does Burbiel 2018: 7), and thus plausibly an intermediate stage of diachronic deaffrication.

affricates may arise, of which by far the most frequent is post-nasal fortition: it is sometimes the case that an excrescent stop arises in the transition from a nasal to a fricative, due presumably to early raising of the velum relative to the release of the oral closure (Ohala 1997; Ohala & Solé 2010). This is shown below in Figure 9 for speakers F01 and F07's productions of the word ⟨егерменче⟩ /jegermenɕe/ 'twentieth', which speaker F07 (right) produces with the excrescent stop.

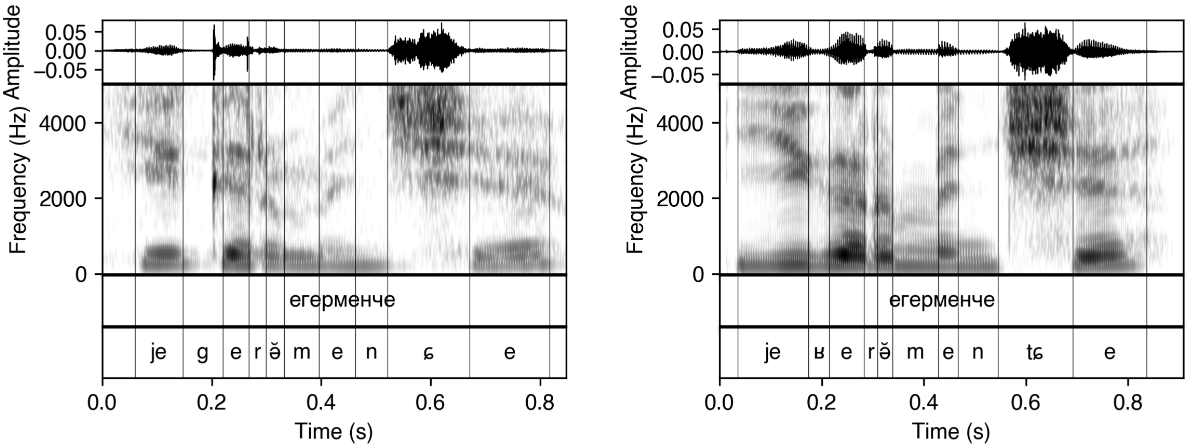


Figure 9: Waveforms and spectra for ⟨егерменче⟩ /jegermenɕe/ 'twentieth' produced by speakers F01 (left) and F07 (right). Note, for F07, the short period of silence corresponding to closure, and the weak plosive-like burst.

Voiceless fricatives

	Mean CoG (Hz)	SD	Tokens
f	4070	1490	49
s	6492	1676	538
ɕ	4296	765	260
ʃ	3712	943	325
χ	2868	872	98
h	2419	771	45
Total			1315

Voiced fricatives

	Mean CoG (Hz)	SD	Tokens
v	1619	727	13
z	5941	1759	596
ʑ	4115	719	48
ʒ	3097	1185	37
ʁ	2145	1070	235
Total			1360

Table 3: Center of gravity (CoG) for all fricatives by voicing, in Hz, with total numbers of tokens shown. Values were calculated from the spectra shown in Figure 11.

Table 3 gives the mean spectral centre of gravity (CoG) with standard deviations for the voiceless and voiced fricatives; this data is visualised in Figure 10, and averaged power spectrum slices are shown in Figure 11. Power spectra were calculated using the middle 50% of the segmented interval for each target fricative (from 25% to 75% of the total duration); very short segments (< 20ms) were excluded from analysis, and spectra were computed using a fixed 20ms

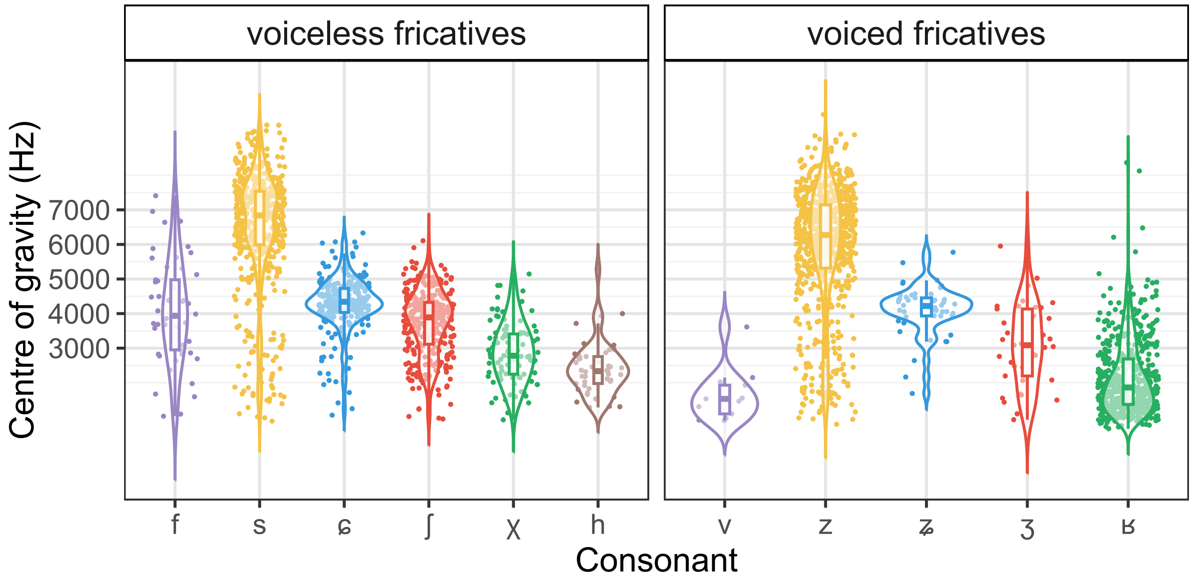


Figure 10: Centre of gravity for the fricatives, showing the distribution of individual tokens, the median (central line), and the 25%—75% interquartile range.

Hamming window applied to the extracted segments. The spectral moments (centre of gravity; also variance, skewness, kurtosis) were then calculated from these spectra, limited to frequencies below 11025 Hz (half the Nyquist frequency) (see e.g. Forrest et al. 1988; Jongman et al. 2000).

Only the post-alveolar /ʃ/ showed a significant effect of front/back vowel context on centre of gravity ($t = -5.72$, $p \ll .0001$, Cohen's $d = 0.698$, with Bonferroni correction for multiple comparisons $\alpha = 0.05/11 = 0.0045$), with front tokens showing higher CoG values than back ones (/ʃ/ back mean 3471 Hz, SD 940 Hz; front mean 4132 Hz, SD 808 Hz, $N = 236$).

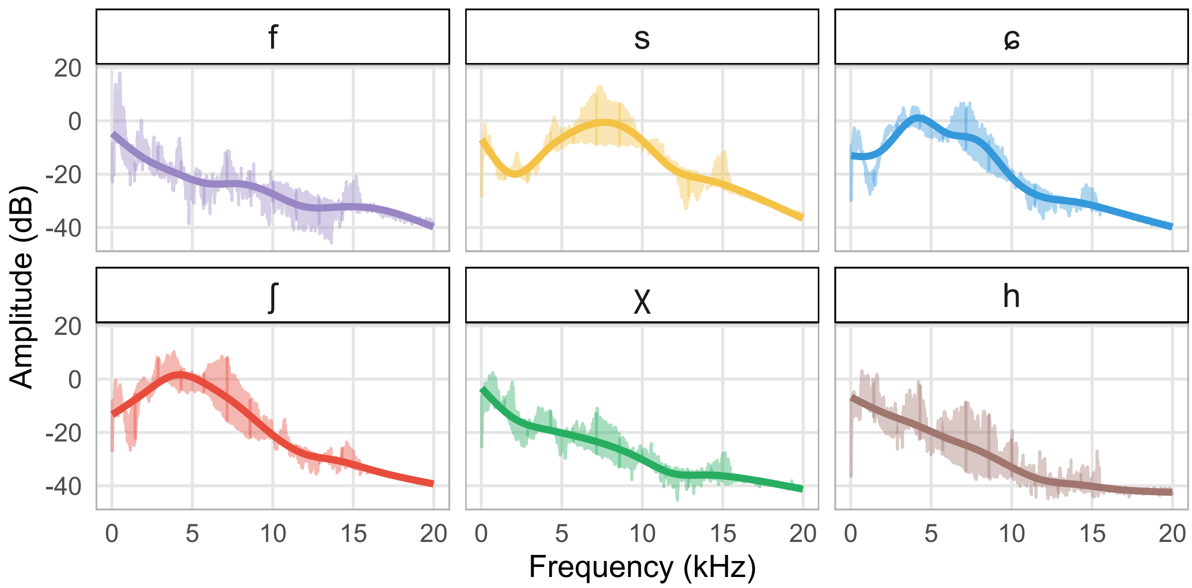


Figure 11: Mean power spectrum slices for voiceless fricatives at six places of articulation. Mean spectra (lighter colors) aggregating all tokens are shown for each fricative, and GAM smoothed spectra (darker) are superimposed.

Predictably, /s z/ are straightforwardly separated from the other fricatives by centre of gravity; within the non-sibilant fricatives, several pairs are not, however, distinguished by CoG alone. The alveolopalatal /ç/ and the post-alveolar /ʃ/ show different means, but substantially overlapping distributions which are not statistically separable. The alveolo-palatals, however, are distinguished by significantly fronted transitions into the following vowel, as illustrated in Figure 12 for the four most frequent vowels /e æ ɑ ə/; this is in agreement with the findings reported in Howson (2015) for the three-way fricative contrast (alveolo-palatal, alveolar, and retroflex) in Lower Sorbian. The voiceless uvular fricative /χ/ and the voiceless glottal fricative /h/ overlapped considerably in values and in spectral shape; as described in the discussion of the consonant inventory above, there was considerable inter- and intra-speaker variation in the distribution of these sounds.

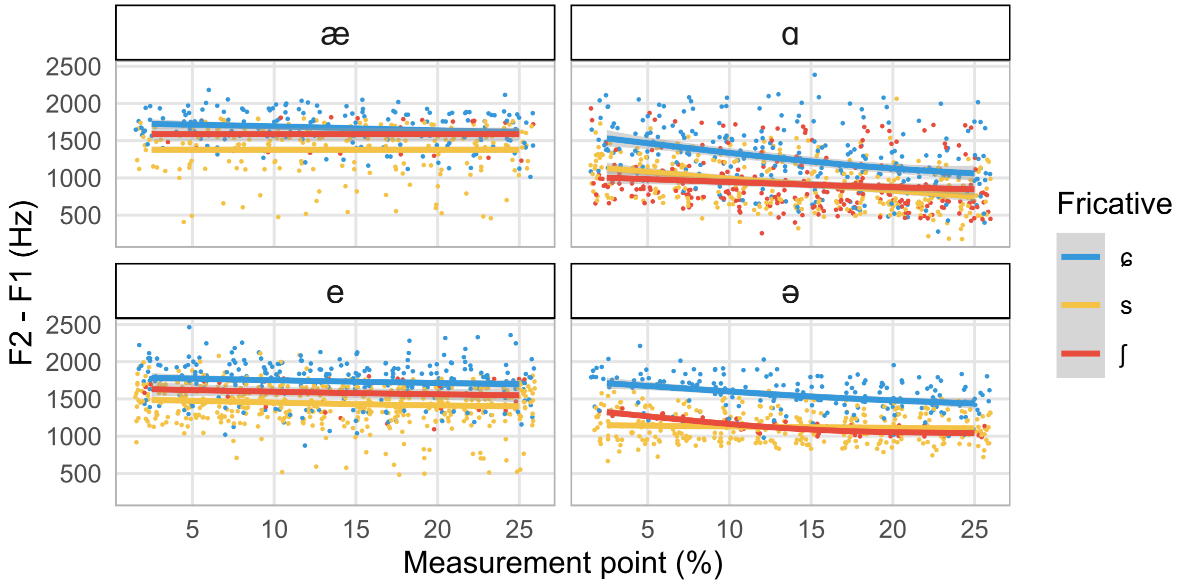


Figure 12: F2-F1 for ten measurements taken during the first 25% of vowel duration, following alveolo-palatal /ç/, alveolar /s/, and postalveolar /ʃ/, shown for four vowels (/e/ $N = 82$, /æ/ $N = 42$, /ɑ/ $N = 96$, /ə/ $N = 58$). Points represent individual measurements and are jittered for visibility; GAM smoothed trajectories are plotted.

Fricative voicing is not robust. Voiced fricatives are not well-tolerated word-initially, even where they arise as allophones of the voiced plosives, and are typically preceded by a prothetic vowel; thus ⟨гаилэлэрендэ⟩ [ŏ.ʁɒ.i.læ.læ.ren.ˈd̪æ] ‘family-PL-POSS.3S-LOC’, ⟨жуылдау⟩ [ə̆ʒu.əɫ.ˈd̪ɑw] ‘buzz-VN’, ⟨зинданнардан⟩ [ĭ.zin.d̪ɑn.nɑ.rə̆.ˈd̪ɑn] ‘jail-PL-ABL’. Figure 13 illustrates this for ⟨зат⟩ [ŏˈzɑt̪] ‘essence, thing’ and ⟨гыйбадэт⟩ [ə.ʁəj.bɑ.ˈðæt̪] ‘worship’. For some speakers, initial /ʁ/ may delete rather than undergoing prothetic repair: thus ⟨гаилэлэрендэ⟩ /ʁɑilæ-lær-en-d̪æ/ ‘family-PL-POSS.3S-LOC’ may be [ɒj.læ.læ.ren.ˈd̪æ] (F03) or [ʔɒʁ.ɒj.læ.læ.ren.ˈd̪æ] (F07), as illustrated in Figure 14. Voicing typically does not persist for the full duration of a voiced fricative, as illustrated in Figure 15 below.

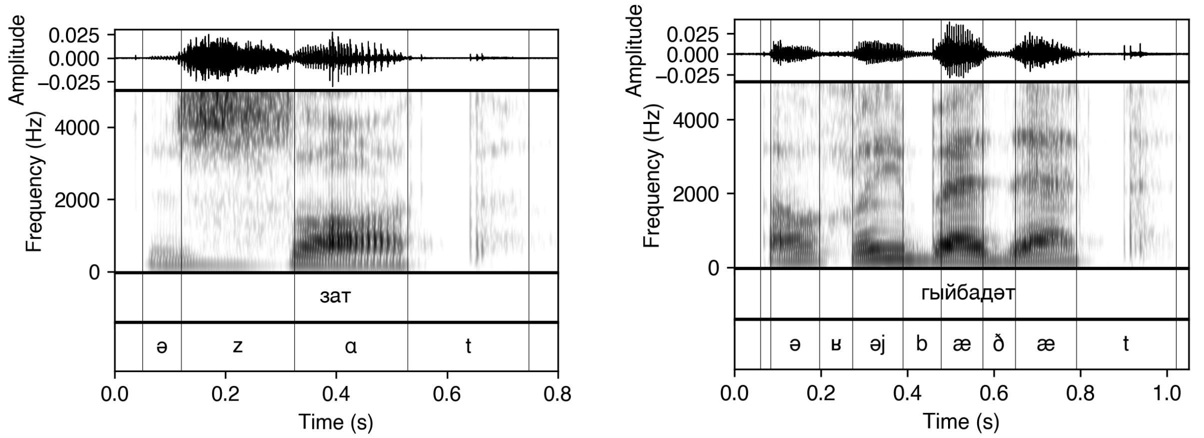


Figure 13: Waveforms and spectra for F07's ⟨зат⟩ [ə̆ˈzɑt̪] 'essence, thing' and ⟨гыйбадəт⟩ [ə.ʁəj.bɑ.ˈðæt̪] 'worship', showing prothetic vowels before word-initial voiced fricatives.

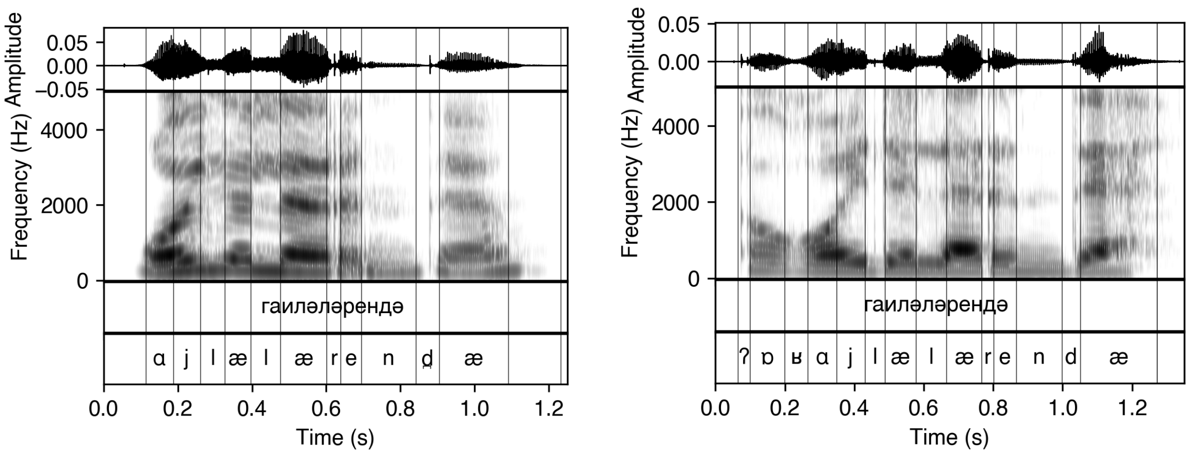


Figure 14: Waveforms and spectra for ⟨гаилəлəрендə⟩ /ʁɑilæ-lær-en-d̪æ/ 'family-PL-POSS.3s-LOC' produced by F03 (left) and F07 (right). For F03, the word-initial fricative deletes entirely [ɒj.læ.læ.ren.ˈd̪æ], for F07 it takes a prothetic vowel that matches the quality of the following root-initial vowel [ʔɒʁ.ɒj.læ.læ.ren.ˈd̪æ].

The voiced fricatives devoice at the phrase level, largely neutralising the voiced-voiceless contrast as for the plosives. Almost all items which were produced in isolation were produced with completely voiceless word-final fricatives, with a few examples of carry-over voicing from the preceding vowel. This is shown in Figure 15. A further illustration is given in Figure 16, for morpheme-final /z/ in the word/utterance-final position in ⟨тəмамлыйбыз⟩ /t̪æmɑmlə-j-bəz/ [t̪æ.mæm.ɬəj.ˈbəz̥] 'finish-PRS-1p, and in word-internal, pre-sonorant position in ⟨тəмамлыйбызмы⟩ /t̪æmɑmlə-j-bəz-mə/ [t̪æ.mæm.ɬəj.bəz.ˈmə] 'finish-PRS-1p-Q'. Unlike the plosives, however, there is a clear voiced-voiceless distinction in word-final and root-final position: intervocalic fricatives never voice at morpheme boundaries, compare e.g. ⟨монысы⟩ [mo.noˈsə] 'this-POSS.3s' and ⟨кызы⟩ [qə.ˈzə] 'girl-POSS.3s'.

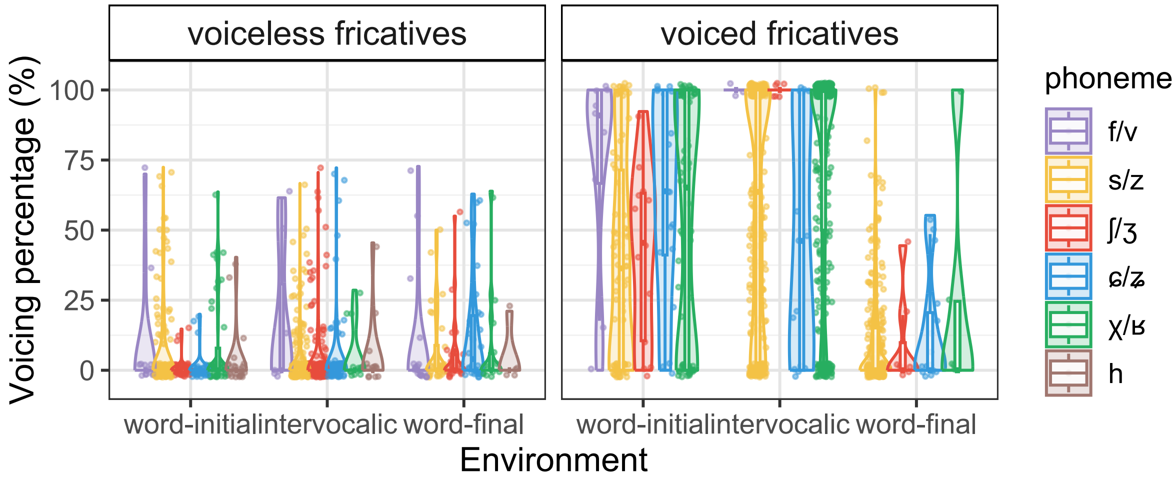


Figure 15: Voicing percentages, calculated automatically as percentage of voiced frames over the entire duration of the segment, for fricatives ($N = 1840$) in words produced in isolation, in word/utterance-initial, intervocalic, and word/utterance-final positions.

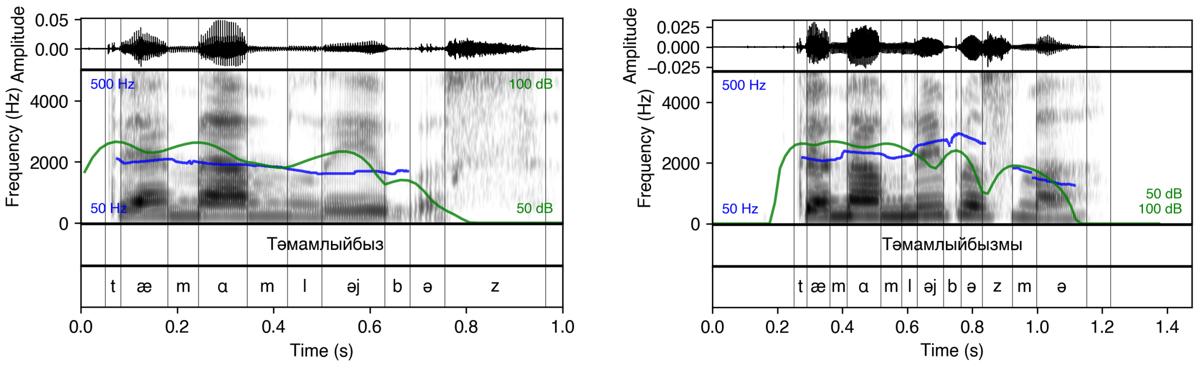


Figure 16: Waveforms and spectra, showing intensity and pitch across the utterance, for ⟨тәмамлыйбыз⟩ [t̪æ.mæm.ɫəj.ˈbəz̥] ‘finish-PRS-1p’ and ⟨тәмамлыйбызмы⟩ [t̪æ.mæm.ɫəj.bəz.ˈmə] ‘finish-PRS-1p-Q’, both produced by speaker F07. Voicing persists from the preceding vowel into a brief portion of the following word-internal /z/, but the word-final /z/ is entirely voiceless.

Although we neglected to elicit examples of this type, we note for completeness that the sibilant fricatives can be subject to voicing assimilation, much as McCollum & Chen (2020) describe for Kazakh. Zakiev (1993: 76) and Şahan Güney (2015: 42) report that morpheme-initial voiceless sibilants trigger regressive devoicing of voiced coda sibilants, ⟨күзсез⟩ /kʉz-sez/ [kʉs.ˈsez] ‘eye-PRIV’.

Nasals

There are three nasals, at the bilabial, dental, and velar places of articulation: /m n ŋ/. The velar nasal cannot occur word-initially. Burbiel (2018: 8) and Şahan Güney (2015: 18) claim that the velar nasal has a uvular realisation [ɴ] in back vowel environments, as has been claimed for Kazakh (Vajda 1994; McCollum & Chen 2020) and for Bashkir (Poppe 1964: 11). Figure 17 shows that for /ŋ/, tokens in back-harmonic words have greater energy in the bands 0–2KHz and 2.5—5KHz than tokens in front-harmonic words, potentially supporting the claim that there are differing realisations in these environments. The effect is small, however, and confirmatory articulatory data is required.

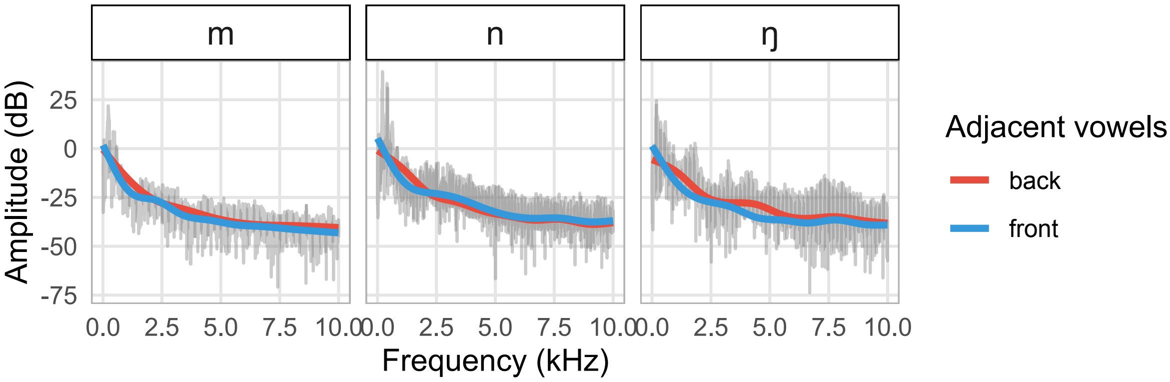


Figure 17: Mean power spectrum slices for nasals ($N = 2480$) at the three nasal places of articulation. Spectral data were extracted from the middle 50% of each nasal (25%-75% of the segment duration), and a Hanning window of this duration was used for the FFT. Mean spectra (grey) aggregating all tokens are shown for each nasal, and GAM smoothed spectra (blue, red) are superimposed for each vowel context condition (front, back).

/n̺/ assimilates to the place of articulation of a following plosive, as in ⟨бүгенгедәй⟩ /bʉgengeda̺æj/ [bʉ.geŋ.ge.ˈd̺æj] ‘today-REL-SIM’ and ⟨хатын-кыз⟩ /χɒt̺ən̺ + qəz/ [χɒ.t̺əŋ‿qəz] ‘womenfolk (lit. woman-girl)’. There is assimilation to the place of some *surface* fricatives, as in ⟨урманга⟩ /urmɑn-ʁɑ/ [ur.mɑŋ.ˈʁɑ] ‘forest-DAT’ (passage), but more usually, epenthesis intervenes between /n̺/ and a following voiced fricative, as in ⟨борынгы⟩ /borənʁə/ [bo.ro.ŋə.ˈʁə] ‘ancient’ [11] . Nasalised vowels are occasional allophones of /n̺/ before voiceless sibilants, as in ⟨икенчесен⟩ /ikenɕesen/ [i.kẽ.ɕe.ˈsen] ‘second-POSS.3S-ACC’ (F07), ⟨буенча⟩ /buenɕɑ/ [bu.jẽ.ˈɕɑ] ‘along; concerning’ (F07). There is typically no assimilation to the place of articulation of a following nasal of different place, as in ⟨ясаганмы⟩ [jɒ.sɒ.ʁɑn.ˈmə] ‘make-PST.3S-Q’.

Lateral

The lateral /l/ is available in all positions; although the native vocabulary generally does not provide examples of word-initial laterals, word-initial laterals in borrowings are well-tolerated and generally surface unrepaired.

[11] Several potential variables intervene here; ⟨борынгы⟩ /borənʁə/ involves a /-ʁə/ or /-gə/ (historically /q/) whose synchronic morphological productivity is unclear. The dative is synchronically very active and alternates between /gæ/ in front environments and /ʁɑ/ in back ones. An analysis of the dative as containing an underlying stop which spirantises is therefore possible, as is an analysis of nasal place assimilation as distinguishing between multiple levels in the morphology. This is difficult to test in this work.

The lateral has an allophonic front-back distribution: if surrounded by back vowels it is clearly and noticeably backed. Both $F2$ and $F1$ during the lateral closure are plausible acoustic correlates of this alternation (see, among others Recasens 2012; Kochetov et al. 2020); the expectation is that $F2$ is lower during the velarised lateral, and $F1$ secondarily higher. Figure 18 illustrates this for 299 tokens produced by the 7 *female* speakers in the sample (values were not normalised); front and back tokens essentially do not overlap in $F1 \times F2$ acoustic space. The effect of front/back context was statistically significant for both $F1$ and $F2$ ($F1$: $t = -9.44$, $p < .001$, Cohen's $d = -0.964$; $F2$: $t = 22.4$, $p < .001$, Cohen's $d = 1.58$, with Bonferroni correction for multiple comparisons $\alpha = 0.05/2 = 0.025$), with front tokens showing lower $F1$ values and higher $F2$ values than back ones: back mean $F1 = 503$ Hz ($SD = 117$) and mean $F2 = 1241$ Hz ($SD = 277$), front mean $F1 = 389$ Hz ($SD = 89.3$) and mean $F2 = 2067$ Hz ($SD = 358$).

For comparison, Nance (2014) gives averages of $F1 = 409$ and $F2 = 1048$ for [ɫ], $F1 = 399$ and $F2 = 1610$ for [l], and $F1 = 359$ and $F2 = 1953$ for [lʲ], for 23 female speakers of Scots Gaelic. Selecting only the 4 female speakers from Kochetov et al. (2020) gives mean $F1 = 451.5$ and mean $F2 = 1316.25$ for [ɫ] and mean $F1 = 406.25$ and $F2 = 2105.25$ for [lʲ] in Kalasha[12]. Comparison to these reference values suggests that the Tatar front and back allophonic variants may best be described as [lʲ] and [ɫ] respectively.

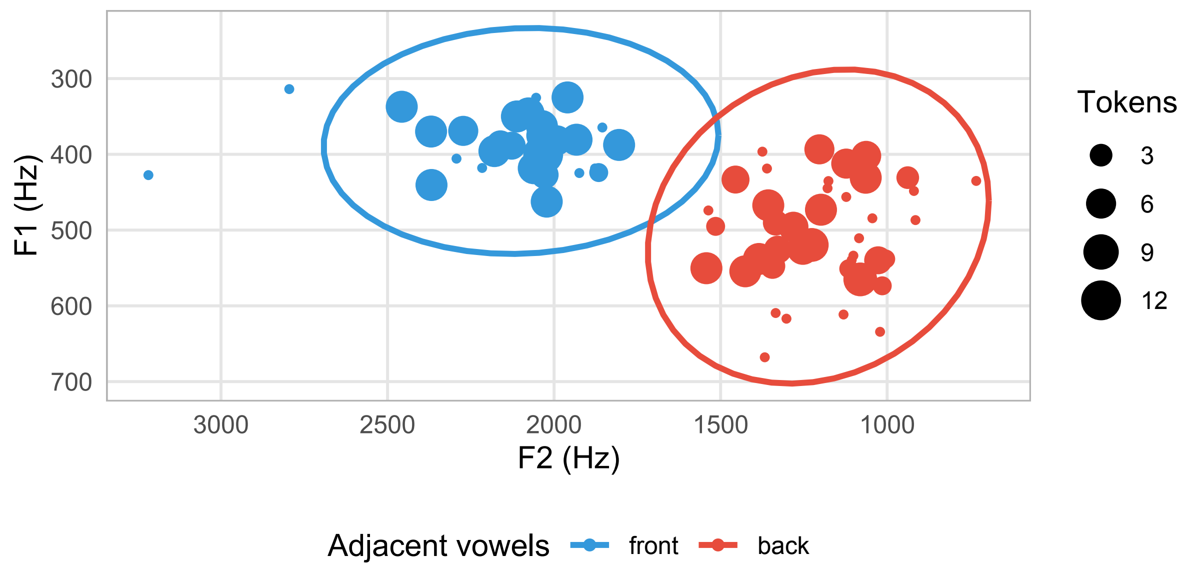


Figure 18: $F1 \times F2$ space for all intervocalic laterals ($N = 299$), coloured by the backness of the surrounding vowels; points are medians aggregated across speakers for each individual word, and ellipses shown represent the spread of the data to 99% confidence.

Affix-initial laterals are subject to some alternations and phonotactic restrictions. Tatar does not generally show the categorical morpho-phonological patterns of obstruentisation (sometimes 'desonorisation') of affix-initial sonorants that are otherwise widespread throughout the Kipchak and Siberian Turkic languages, and that cause affix-initial sonorants to surface as plosives in

[12] Values computed by us from their table VII, p. 3026, to exclude male speakers for comparability.

many of these languages, including the nearby Bashkir (see McCollum & Chen 2020 for Kazakh; for detailed overviews, see, among others, Gouskova 2004; Washington 2010 and the survey in Gopal 2018: ch. 2); affix-initial sonorants are typically consistently sonorant. However, some phonetic variation arises, which suggests a similarity between contemporary Tatar and the prior states of the other languages in which the phonologised alternation is seen: Figure 19 shows two tokens of [t-l] for speaker M01, ⟨асатлык⟩ /ɑsɑt̪-ləq/ [ɒ.sɒt̪.ˈɬəq] 'peaceful-NMLZ', in which this sequence is rendered a clear period of voiceless frication corresponding to the lateral, and ⟨дәүләтләрнең⟩ /d̪æwlæt̪-lær-neŋ/ [d̪æw.læ.t̪ːær.ˈneŋ] 'state-PL-GEN' in which there is instead a single prolonged stop closure. Almost all speakers produced ⟨кабатлады⟩ 'repeat-PST.3S' in the carrier sentence ⟨Ул [X] сүзен кабатлады⟩ /ul [X] sʉzen qɑbɑt̪lɑd̪ə/ with [tː], as [qɒ.bɒ.t̪ːɑ.ˈd̪ə]. Unlike the situation in the related languages, variants of this type arise only when the preceding voiceless obstruent is homorganic with the lateral.

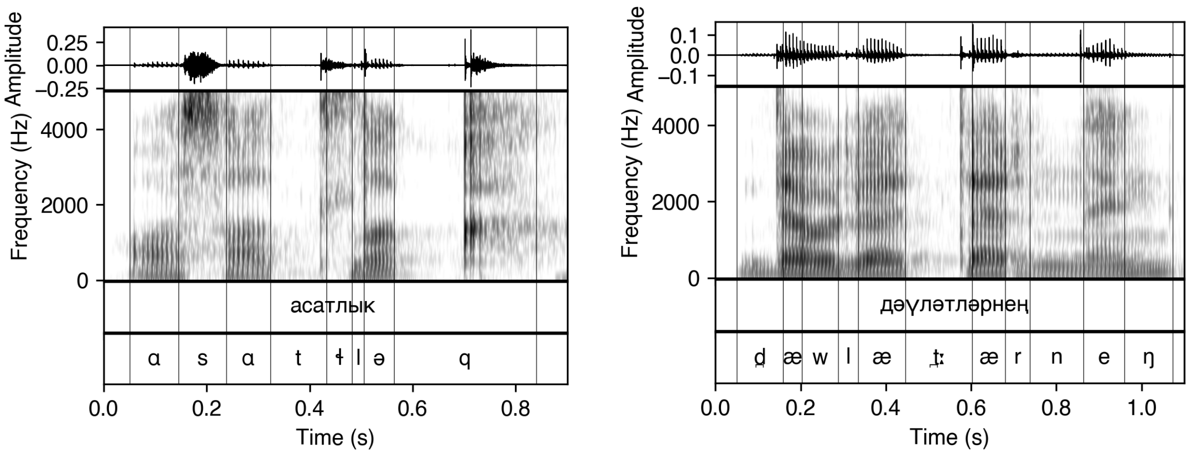


Figure 19: Waveforms and spectra for ⟨асатлык⟩ /ɑsɑt̪-ləq/ [ɒ.sɒt̪.ˈɬəq] 'peaceful-NMLZ' and ⟨дәүләтләрнең⟩ /d̪æwlæt̪-lær-neŋ/ [d̪æw.læt̪.tær.ˈneŋ] 'state-PL-GEN', both produced by speaker M01.

Affix-initial laterals nasalise if preceded by a root-final nasal (unlike the case of the affix-initial stops above, this applies even in the absence of a following nasal coda): compare, for the plural /-l(æ/ɑ)r/ ⟨базарларда⟩ [bɒ.zɒr.ɫɑr.ˈd̪ɑ] 'market-PL-LOC' and ⟨балаларга⟩ [bɒ.ɫɒ.ɫɑr.ˈʁɑ] 'child-PL-DAT' but ⟨заманнарда⟩ [zɒ.mɒn.nɑr.ˈd̪ɑ] 'time-PL-LOC' and ⟨адымнарын⟩ [ɒ.d̪əm.nɑ.ˈrən] 'step-PL-POSS.3S-ACC'.

Trill or tap

The rhotic /r/ is typically a trill, as in ⟨вәзирлек⟩ [væ.zir.ˈlek] 'position of vizier', although often realised as a tap [ɾ] (that is, with a single contact) and variably devoiced in some word-final, pre-consonantal, and phrase-final positions; like the obstruents, the trill is invariably voiceless phrase-finally. It does not occur word-initially in the native vocabulary, but can be found in a few borrowings: ⟨рух⟩ [ruχ] 'soul', ⟨рус⟩ [rus] 'Russian'. However, like the voiced fricatives (and largely unlike the lateral), these word-initial rhotics are often preceded by a prothetic vowel: ⟨рижа⟩ [ə̆.ri.ˈʑɑ] ‘hope’ (F07), ⟨рәхәтләнеп⟩ [ə̆.ræ.χæ.t̪ːæ.ˈnep] ‘take.pleasure.in-VADV’ (F07), ⟨рус⟩ [ŭ.rus] ‘Russian’ (F06). This type of repair is also reported for Kazakh (McCollum & Chen 2020), but in Tatar unites the rhotic and the voiced fricatives to the exclusion of other sounds.

As with the other sonorants, realisations of the trill vary on the basis of the backness of the surrounding vocalic environment: since formant structure is not consistently reliable, unlike in the lateral above, Figure 20 illustrates, with averaged power spectra, that front realisations of the trill have more energy in higher-frequency bands than do back realisations.

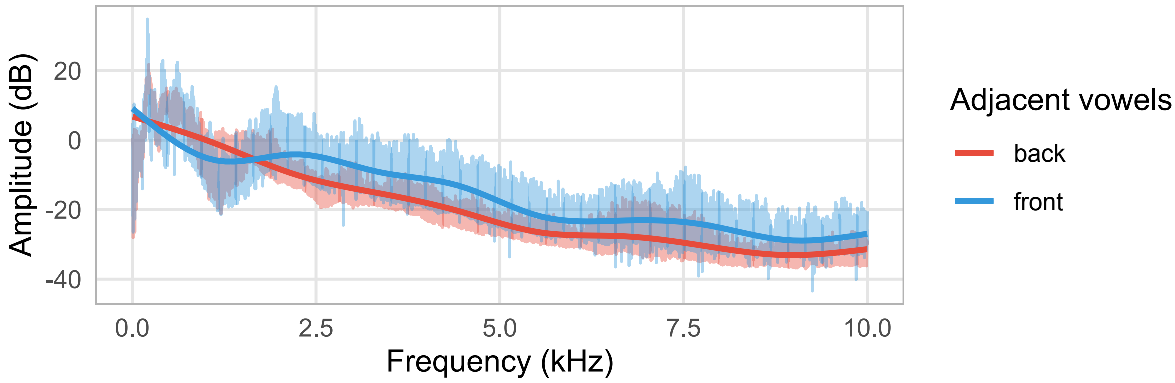


Figure 20: Mean power spectrum slices for intervocalic trills (N = 1071) in front and back environments. Spectral data were extracted from the middle 50% of each token (25%-75% of the segment duration), and a Hanning window of this duration was used for the FFT. Mean spectra aggregating all tokens are shown for each condition (blue/front, red/back), with GAM smooths superimposed.

Epenthetic vowels often intervene both between a preceding coda consonant and following onset rhotic across a syllable boundary, as in ⟨кебегрәк⟩ /kebek-ræk/ [ke.be.ge.ˈræk] ‘similar.to-COMP’ for all speakers, and between a coda rhotic and another following onset consonant: ⟨аерылырга⟩ /ɑjerəl-ər-ʁɑ/ [ɒ.je.rə.ɫə.rə.ˈʁɑ] ‘be.separated-VADJ-DAT’.

Adjacent to a sibilant fricative, the rhotic is typically devoiced and fricated: ⟨парчалау⟩ [pɒr̝̊.ɕɒ.ˈɫɑw] ‘fragment-VN’, ⟨нәрсәседер⟩ [nær̝̊.sæ.se.ˈd̪er] ‘something-POSS.3S-COP’, ⟨меңнәрчә⟩ [meŋ.nær̝̊.ˈɕæ] ‘thousand-PL-EQ’ (all speakers); devoicing may also occur with voiceless obstruents more generally, as in ⟨куркак⟩ [qur̥.ˈqɑq] ‘coward’ (F07, passage), but these realisations are typically devoiced trills rather than truly fricated, as illustrated in Figure 21. Very occasionally, fricated realisations arise even when the following obstruent is voiced, as in ⟨баганаларда⟩ [bɒ.ʁɒ.nɑ.ɫɑr̝̊.ˈd̪ɑ] ‘column-PL-LOC’ (F05).

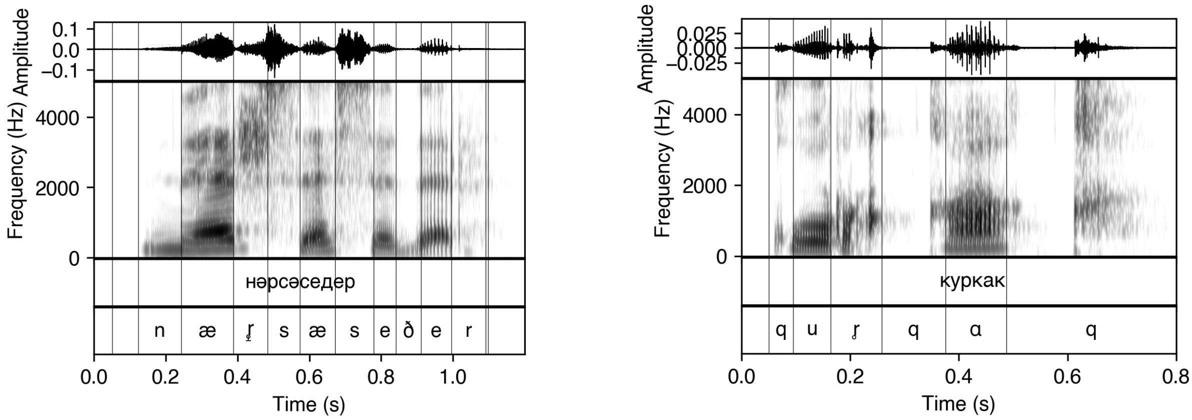


Figure 21: Waveforms and spectra for ⟨нәрсәседер⟩ [næɾ̥.sæ.se.ˈd̪er] ‘something-POSS.3s-COP’ and ⟨куркак⟩ [quɾ̥.ˈqɑq] ‘coward’, both produced by speaker F07.

Other approximants

There are two non-lateral non-rhotic approximants[13] /w j/ at the labial and palatal places of articulation. These, especially /w/, are not consistently differentiated from the vowels in the orthography: orthographic ⟨у⟩ may represent /u/, /w/, or glide-vowel sequences like /uw/, and ⟨ү⟩ may represent /ʉ/, /w/ and /ʉw/. We illustrate the non-syllabic status of /w/ with the examples in Figure 22, showing the words ⟨башлауга⟩ /bɒʃ.ɫɒw.ˈʁɑ/ ‘begin-VN-DAT’ and ⟨көлүе⟩ [kø̞.lʉ.ˈwe] ‘laugh-VN-POSS.3s’ produced by speaker F07 in a syllabification task[14]: /w/ is invariably syllabified outside the nucleus, and is clearly lower-intensity than the adjacent vowels (the number of intensity peaks corresponds to the number of true vowels throughout).

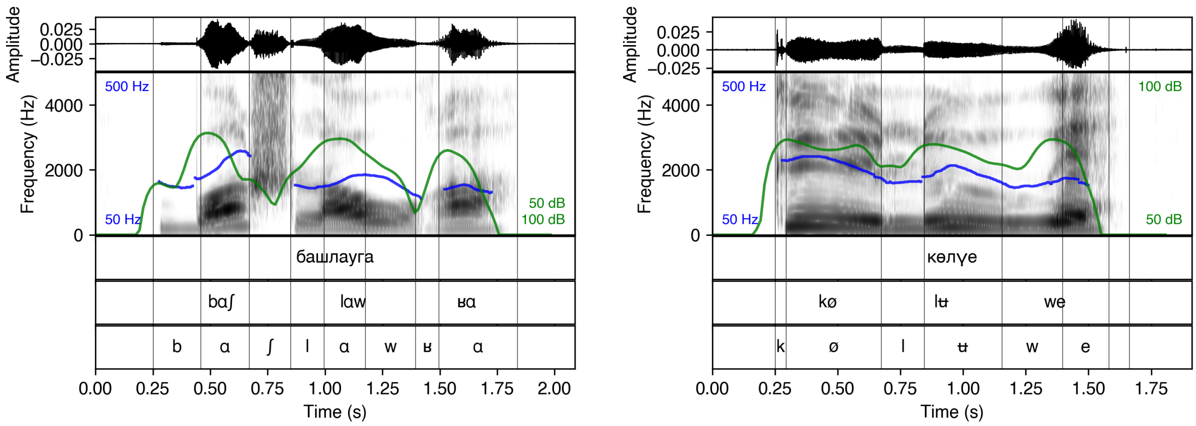


Figure 22: Waveforms and spectra for /w/ in ⟨башлауга⟩ /bɒʃ.ɫɒw.ˈʁɑ/ ‘begin-VN-DAT’ and ⟨көлүе⟩ [kø̞.lʉ.ˈwe] ‘laugh-VN-POSS.3s’ produced by speaker F07.

Word-initially, /j/ can fricate substantially ([ʝ]), ⟨ераклыкка⟩ [ʝə.rɑq.ɫəq.ˈqɑ] ‘far-NMLZ-DAT’ (F01), ⟨юлына⟩ /ju.ɫə.ˈnɑ/ [ʝu.ɫə.ˈnɑ] ‘way-POSS.3s-DAT’ (F07), and /w/ may begin with

[13]The descriptive literature typically uses the term ‘semi-vowel’, as e.g. Poppe (1963: 13).

[14]The task instructions given were (Russian) ⟨Произносите по слогам (например ба-ла-лар)⟩, lit. ‘Pronounce by syllable (for example ba-la-lar)’

a prolonged vowel-like steady state [uw], ⟨вə⟩ [uwæ] 'and' (F07); both cases are illustrated in Figure 23 below. The glide can also fricate and devoice word-finally, as in ⟨тѳтенедэй⟩ (F03, F06) [t̪ø.t̪ø.nø.ˈd̪æj] 'smoke-POSS.3S-SIM'.

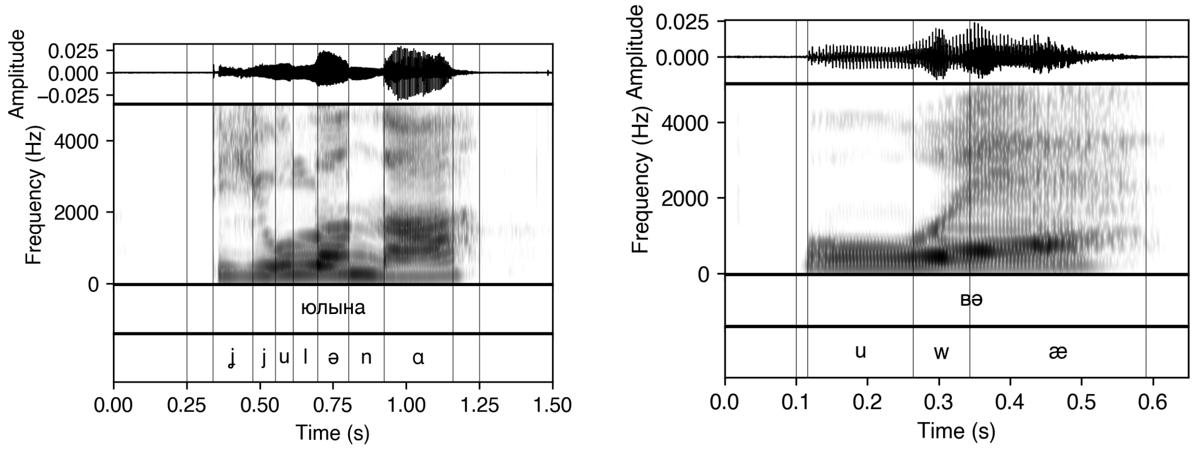


Figure 23: Waveforms and spectra for ⟨юлына⟩ [j̥u.ɬə.ˈnɑ] 'way-POSS.3S-DAT' (left) and ⟨вə⟩ [uwæ] 'and' (right), produced by speaker F07, showing the frication of initial /j/ and the steady-state period in initial /w/.

The glides are coarticulatorily fronted and backed in the respective vocalic environments (Burbiel 2018: 6), illustrated in Figure 24.

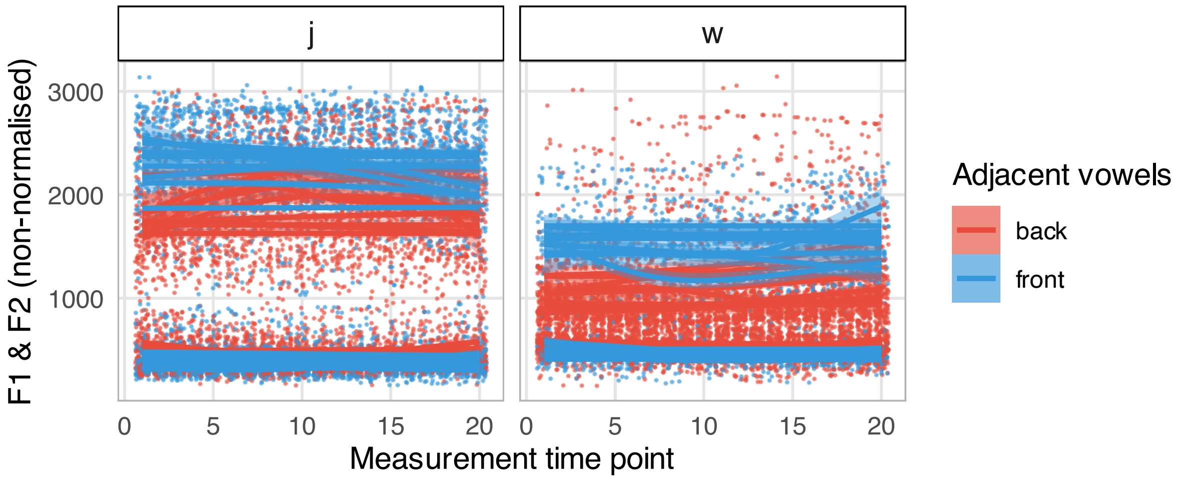
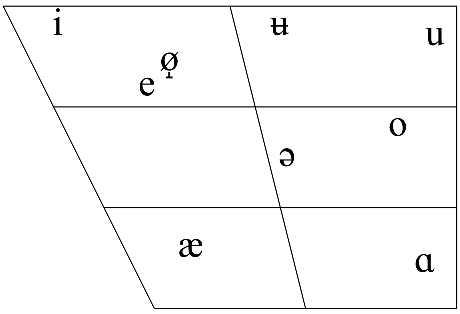


Figure 24: $F1$ and $F2$ space measured at 20 points during the full duration of each glide /j/ /w/ in front- or back-harmonic words ($N = 502$), coloured by the backness of the surrounding vowels; lines are GAM trajectories aggregated across words for each individual speaker, and points shown represent individual measurements (jittered horizontally for visibility).

Vowels

Previous descriptive work on the standard variety of Tatar has converged on an inventory of either nine (Poppe 1968: 8; Zakiev 1993: 59–60, 89–90; Şahan Güney 2015: 19–20; Burbiel 2018: 1-2) or ten (Bogoroditskiĭ 1934: 27–28; Bogoroditskiĭ 1953: 82–83; Faseev 1966: 814; Kurbatov et al. 1969: 82–86; Baskakov 1988: 136–137; Comrie 1997: 900–901; Conklin & Dmitrieva 2019: 1555)[15] vowels. All proposed systems differ in internal structure to some degree, with an essential tension in transcription lying in the retention of front-back symmetry (motivated by the pervasive harmony system) versus the representation of centralisation in acoustic space. Tatar has undergone a series of vowel shifts that have profoundly affected its phonology in comparison to the other Turkic languages' (as has the neighbouring Bashkir); historical overviews appear in Hattori (1979–1980) and Berta (1982).

Here we propose an inventory of 9 vowel phonemes /i ʉ u e ə ø̝ o æ ɑ/, illustrated below. Positions in the chart roughly reflect the position of the vowel in $F1 \times F2$ space (Figure 25). The choice of symbols is largely identical to that given in Conklin & Dmitrieva (2019), save the absence of a tenth vowel and the indication of raising in /ø̝/ (for which Conklin & Dmitrieva (2019) have plain /ø/).



æ	⟨тәм⟩	/t̪æm/	taste
ɑ	⟨таз⟩	/t̪ɑz/	bald
e	⟨тез⟩	/t̪ez/	knee
ə	⟨тыш⟩	/t̪əʃ/	exterior
i	⟨тиз⟩	/t̪iz/	rapid
o	⟨тоз⟩	/t̪oz/	salt
ø̝	⟨төз⟩	/t̪ø̝z/	proper
u	⟨туй⟩	/t̪uj/	wedding
ʉ	⟨түл⟩	/t̪ʉl/	ovary

Table 4 gives sample Hertz values[16] for two speakers averaged over all tokens, and Fig-

[15] Harrison & Kaun (2003) propose a nine-vowel system for the Tatar variety of Namangan, Uzbekistan.
[16] Comparable values are given by Zakiev (1993: 65) and by Conklin & Dmitrieva (2019).

ures 25 and 26 show distributions of these vowels across all speakers in normalised $F1 \times F2$ space. For each vowel token, measurements of $F1$, $F2$, and $F3$ were made at 25%, 50%, and 75% of the duration of the vowel.

Table 4: Mean $F1$, $F2$, and $F3$ in initial syllables, for two speakers, M01 and F07.

	Male speaker (M01)			Female speaker (F07)		
	$F1$ (SD)	$F2$ (SD)	$F3$ (SD)	$F1$ (SD)	$F2$ (SD)	$F3$ (SD)
æ	617 (±98.5)	1889 (±177)	2895 (±164)	589 (±107)	2014 (±282)	2955 (±587)
ɑ	615 (±87.9)	1140 (±219)	2720 (±406)	639 (±83.7)	1233 (±167)	2750 (±259)
e	439 (±73.5)	1918 (±235)	2731 (±387)	438 (±36.9)	2067 (±121)	3008 (±288)
ə	503 (±73.4)	1486 (±254)	2845 (±267)	522 (±79.5)	1588 (±246)	2974 (±409)
i	280 (±47.9)	2575 (±290)	3494 (±446)	307 (±67.0)	2298 (±330)	3175 (±446)
o	500 (±103)	1278 (±200)	2791 (±487)	496 (±105)	1306 (±194)	2739 (±277)
ø̞	365 (±77.0)	2033 (±331)	2732 (±256)	388 (±54.8)	2013 (±146)	2771 (±260)
u	411 (±47.8)	967 (±276)	2774 (±387)	392 (±63.7)	1039 (±157)	2789 (±322)
ʉ	335 (±30.9)	1327 (±278)	2295 (±225)	333 (±70.7)	1516 (±274)	2732 (±205)

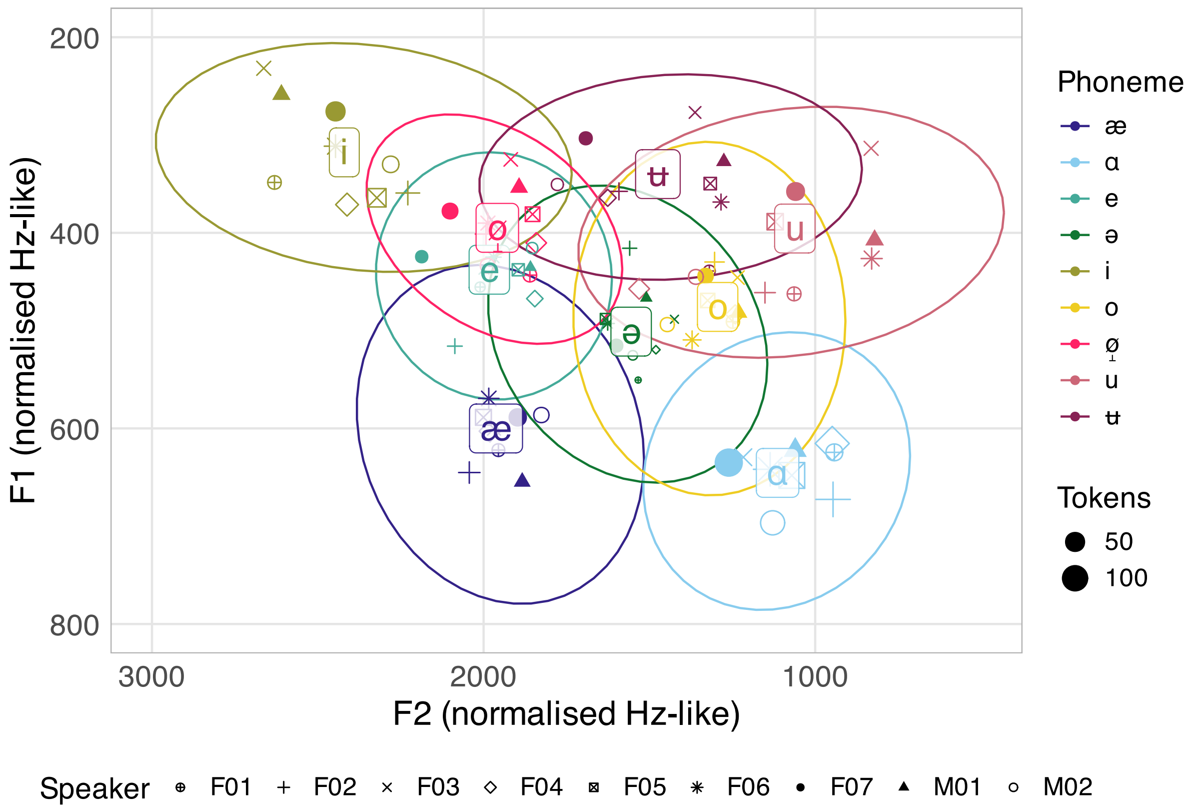


Figure 25: Plot of $F1$ and $F2$ values of 9 Tatar vowels in initial syllables ($N = 2444$, of which 405 monosyllables) produced by all speakers, measured at the midpoint. Values were z-score normalised and rescaled to Hz-like; ellipses show the spread of all $F1$ and $F2$ values to 75% confidence, and text labels are located at the centroid for the category. Points correspond to the median $F1$ and $F2$ for each speaker.

Values that appear in Table 4 are non-normalised averages of the midpoint measurement over all tokens for two of the speakers in our sample, representing typical male and female values. The values appearing above in Figure 25 were z-score normalised for presentation and scaled to Hz-like values as $F_i = \bar{F}_i + F_i^N \times \sigma(F_i)$[17]. Figure 25 shows considerable overlap in the realisations of several of these vowels in $F1 \times F2$ space; this is in accordance with previous reports (Conklin & Dmitrieva 2019) of crowding of the central vowel space. As Figure 26 shows, this persists across vowels in all positions, but the phonotactics of the language heavily restrict the set of possible vowels outside the initial syllable.

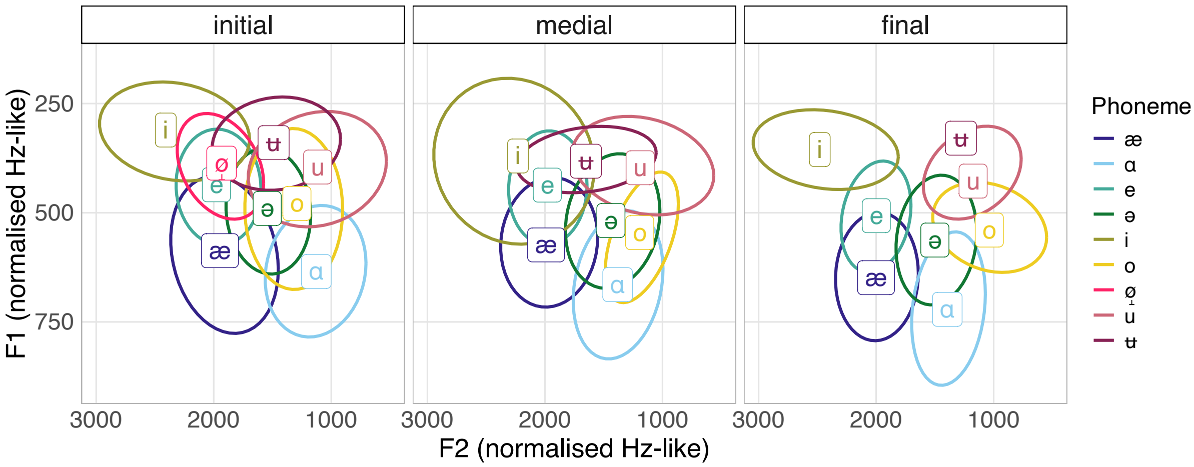


Figure 26: Plot of $F1$ and $F2$ values of 9 vowels in polysyllabic words ($N = 7003$), by position initial/medial/final, showing 75% confidence ellipses save where too few tokens are available.

As Table 4 shows, $F3$ alone is not a wholly reliable acoustic correlate of the rounding contrast in Tatar; very similar facts have been reported for the Kipchak languages Crimean Tatar (McCollum & Kavitskaya 2022) and Kazakh (McCollum 2015), and for the more distant Dolgan (Gopal et al. 2025). Figure 27 illustrates further for reference values in initial syllables only: $F2$ reliably indicates the contrast between /i/ and /ʉ/ ($t(324) = 24.65$, $p < .001$) and between /ə/ and /o/ ($t(212) = 7.18$, $p < .001$), but not /e/ and /ø̞/ ($t(272) = -0.31$, $p = .759$); $F3$ reliably distinguishes /i/ and /ʉ/ ($t(341) = 14.68$, $p < .001$), and /e/ and /ø̞/ ($t(227) = 2.82$, $p = .005$), but not /ə/ and /o/ ($t(201) = 0.88$, $p = .380$). $F3 - F2$ difference significantly distinguishes /e/ and /ø̞/ ($t(261) = 3.29$, $p = .001$) and /ə/ and /o/ ($t(206) = -2.81$, $p = .005$), but not /i/ and /ʉ/.

The status of the putative tenth vowel, orthographically the digraph ⟨ый⟩, is disputed, and is the largest source of disagreement between previous descriptions; for examples, compare ⟨абый⟩ 'older brother' and ⟨абыйсы⟩ 'older.brother-POSS.3S'. It is analysed by (among others) Berta (1998: 283); Berta (2022: 303) as a vowel-glide sequence /əj/; in contrast, it is analysed by Comrie (1997: 900–901) and Conklin & Dmitrieva (2019: 1555) as a monophthongal /ɨ/ which diphthongises in stressed positions, with Comrie making the further argument that sym-

[17] F_i^N is the z-score value for formant i, $\bar{F}_i$ the mean F_i and $\sigma(F_i)$ the standard deviation in F_i over all tokens.

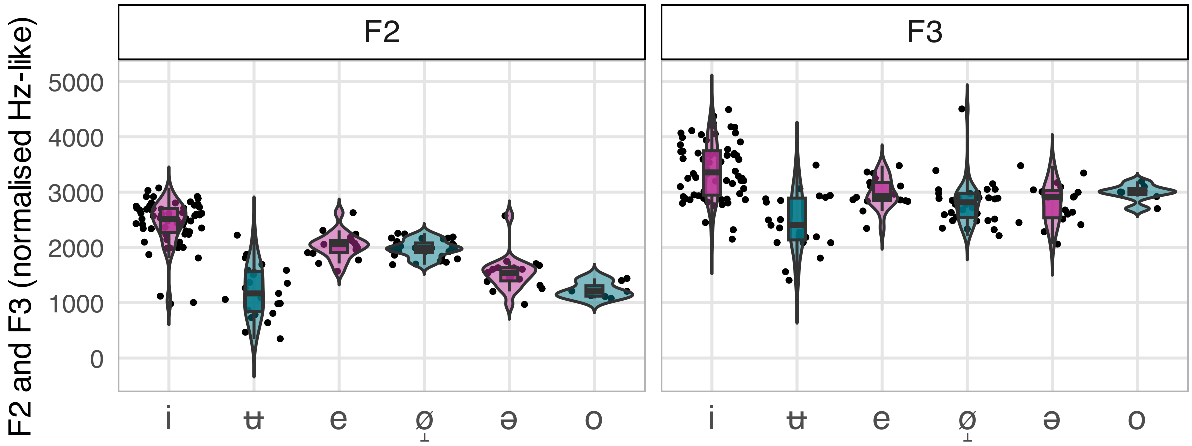


Figure 27: Distributions of *F*2 and *F*3 values (normalised) in monosyllables and initial syllables (*N* = 1036) for 6 of 9 Tatar vowels, arranged by presumed paired status in the phonological system: i–ʉ, e–ø̞, ə–o, and showing the distribution of individual tokens, the median (central line), and the 25%—75% interquartile range.

metric diphthongisation occurs in front /i/, and that as such there should be understood to be two monophthongs which constitute a front-back pair. From the point of view of the diachrony, sequences of this type derive from historic VC sequences in words of Turkic and Mongolic origin. Elsewhere, orthographic ⟨ый⟩ is conventionally used to render any of Arabic /iʔ/, /ʔi/, or /iː/, but it is not clear how these were adapted into Tatar at the original date of borrowing, or on what basis (i.e. with access to Arabic phonetic input, or via an intermediary language like Persian, or purely orthographically), and the orthography significantly postdates this event. With respect to the native vocabulary, the straightforward assumption is that a gradual lenition of the C has occurred, whether with or without subsequent monophthongisation to /ɨ/; a parallel development gives rise to /i/ (Comrie 1997: 901), which is more uncontroversially taken to be monophthongal by all authors. In the analysis of Conklin & Dmitrieva (2019: 1558), the former monophthongisation has run to completion, and ⟨ый⟩ is a monophthongal /ɨ/ that undergoes synchronic diphthongisation to [ɨj] when stressed (in their account, /i/ is subject to similar synchronic diphthongisation); acoustic evidence that ⟨ый⟩ shows monophthong-typical *F*2 in unstressed syllables is given by Conklin & Dmitrieva for their corpus of speakers.

We do not replicate this finding for all our speakers (although examine M02 in Figure 28), and therefore suggest that the more conservative state still persists within the population, and that ⟨ый⟩ continues to represent an underlying vowel-glide sequence /əj/ in which the glide may reduce or even delete entirely in unstressed syllables. We find Vj realisations of ⟨ый⟩ in both stressed and unstressed syllables, illustrated with *F*2 measurements in Figure 28: stressed and unstressed /əj/ differ in degree of *F*2 change from beginning to end, but most are clearly not monophthongal. /əj/-monophthongisation is very occasionally represented in our dataset to a limited extent, largely in connected speech: see the recorded passage, ⟨шушындый куркак кеше⟩ [ʃuʃundɨ qurqɑq kʃe]‘such (a) coward(ly) person’.

These trajectories were analysed using generalised additive models (GAMs) implemented in R (`mgcv`; Wood 2011), with time represented using normalised measurement points 1–20. Models included speaker-specific random intercepts to control for inter-speaker variation. Significant F2 increases from onset to offset were seen for both stressed (+606.2 Hz, t(57), $p \ll .00001$) and unstressed (+389.1 Hz, t(32), $p \ll .00001$) intervals; F1 decrease was statistically significant for stressed intervals (–74.6 Hz, t(57), $p = 0.003$), but not for unstressed ones (+13.7 Hz, $p = 0.67$). Stressed /əj/ was typically longer (stressed mean duration = 198ms, SD = 73.4ms; unstressed mean duration = 106ms, SD = 19.4ms). Neither our study ($N = 91$, 9 speakers) nor Conklin & Dmitrieva's ($N = 209$, 27 speakers), however, is based on a very large number of tokens; as such, it is hoped that future work on the language will provide a more conclusive answer.

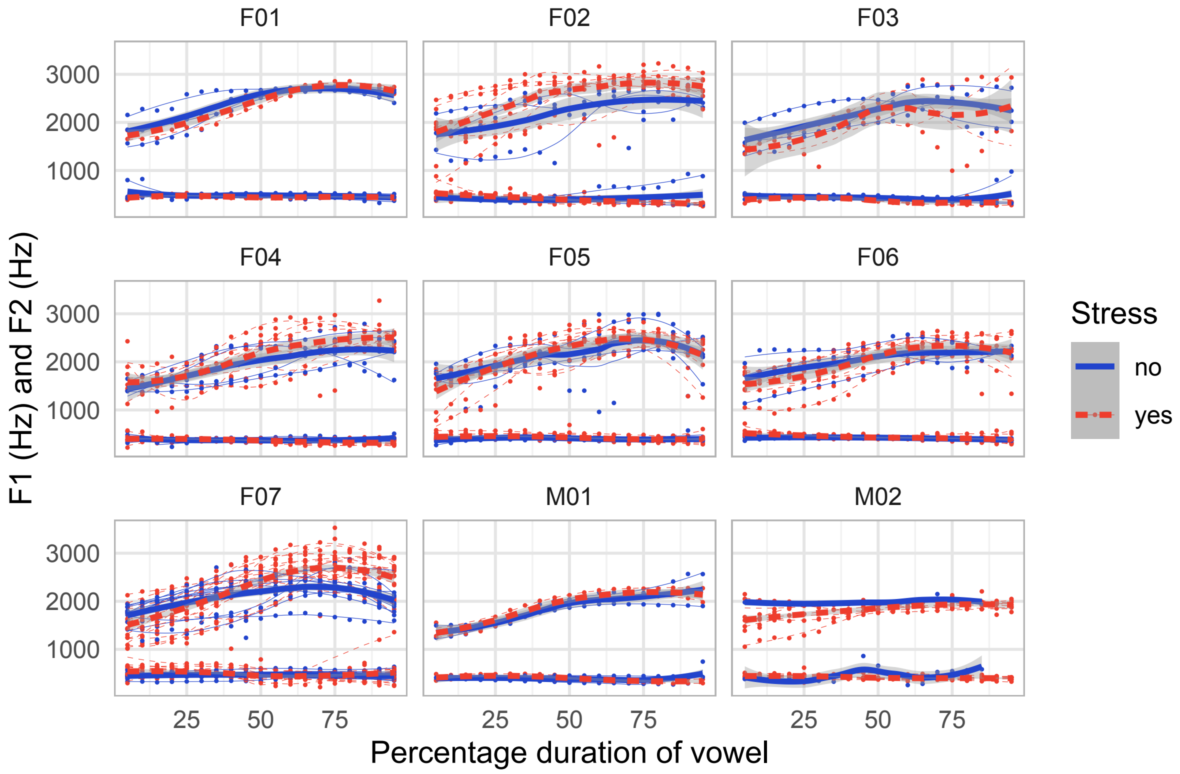


Figure 28: F1 and $F2$ for the glide-vowel sequence ⟨ый⟩ /əj/ against timestamp, measured at 5% intervals from 5% to 95% of the duration of the vowel, grouped by speaker and coloured by stress; for each speaker, aggregated GAMs are shown over all tokens, and individual measurement points are marked and grouped by token. This plot represents 90 total tokens: for speakers F01, F03, and M01, $N = 5$ (3 stressed, 2 unstressed); for speakers F02, F04, F06, and M02 $N = 12$ (4 unstressed, 8 stressed); for speaker F07 $N = 27$ (10 unstressed, 17 stressed).

Low vowels

There are two low vowels, a front, open-mid /æ/ and a back, open /ɑ/. /æ/ is often the non-IPA ⟨ä⟩ in the descriptive literature (Poppe 1968: 9, Comrie 1997: 900, Şahan Güney 2015: 22, Berta 2022: 303), although one source (Berta 1998: 283) uses ⟨e⟩. /ɑ/ is typically ⟨a⟩ in the literature (Poppe 1968: 9, Comrie 1997: 900, Şahan Güney 2015: 22, Berta 2022: 303).

/æ/ is front and unrounded for all speakers. It is higher ([ɛ]) in unstressed initial and medial syllables than in monosyllables and final syllables; compare the two /æ/s in ⟨кесәсеннән⟩ [kĕ.sɛ.sen.ˈnæn] 'pocket-POSS.3S-ABL'. This can be seen over all data in Figure 26, and is further illustrated in Figure 29.

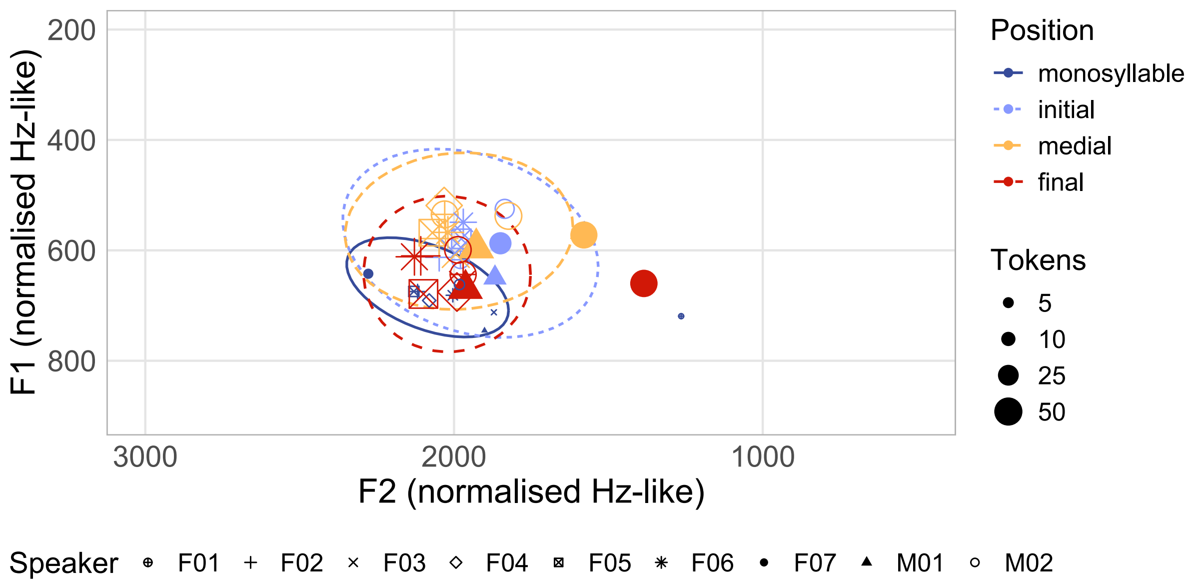


Figure 29: $F1 \times F2$ space for /æ/ ($N = 1104$) by position monosyllable/initial/medial/final. [æ] appears in stressed syllables, that is, in monosyllables and the final syllables of polysyllabic words; tokens in unstressed syllables are higher. Points show aggregated medians for each speaker, and ellipses show the spread of the data (75% confidence).

/ɑ/ is categorically rounded [ɒ] in initial syllables (Poppe 1968: 10, Kurbatov et al. 1969: 83, Comrie 1997: 900, Şahan Güney 2015: 22, Conklin & Dmitrieva 2019). Figure 30 shows this as movement in $F1 \times F2$ for initial (and monosyllable) and final tokens of /ɑ/; word-final tokens have higher F2 and F1 than word-initial tokens and those in monosyllables, consistent with the expected effect of lip rounding on the latter groups.

This rounding can have a gradient coarticulatory effect on /ɑ/ in following syllables, which drops off rapidly after the second syllable; gradualness in this respect is reported by Comrie (1997: 900), by Burbiel (2018: 2) and by Kurbatov et al. (1969: 83), with the latter also reporting an impressionistic description of greater labialisation in the vicinity of labial and uvular consonants. An illustration appears in Figure 31, which shows *non-normalised* tokens of /ɑ/ in 3 Cɑ.Cɑ(.Cɑ)(.Cɑ) words of two, three, and four syllables, ⟨бала⟩ /bɑlɑ/

[bɒ.ˈɫɑ] ‘child’, ⟨балалар⟩ /bɑlɑlɑr/ [bɒ.ɫɒ.ˈɫɑr] ‘child-PL’, and ⟨балаларга⟩ /bɑlɑlɑrʁɑ/ [bɒ.ɫɒ.ɫɑr.ˈʁɑ] ‘child-PL-DAT’, produced by female speakers ($N = 73$ vowels measured, including repetitions). $F1$ and $F2$[18] diverge both from vowel to vowel, and within the second vowel of the two-syllable ⟨бала⟩.

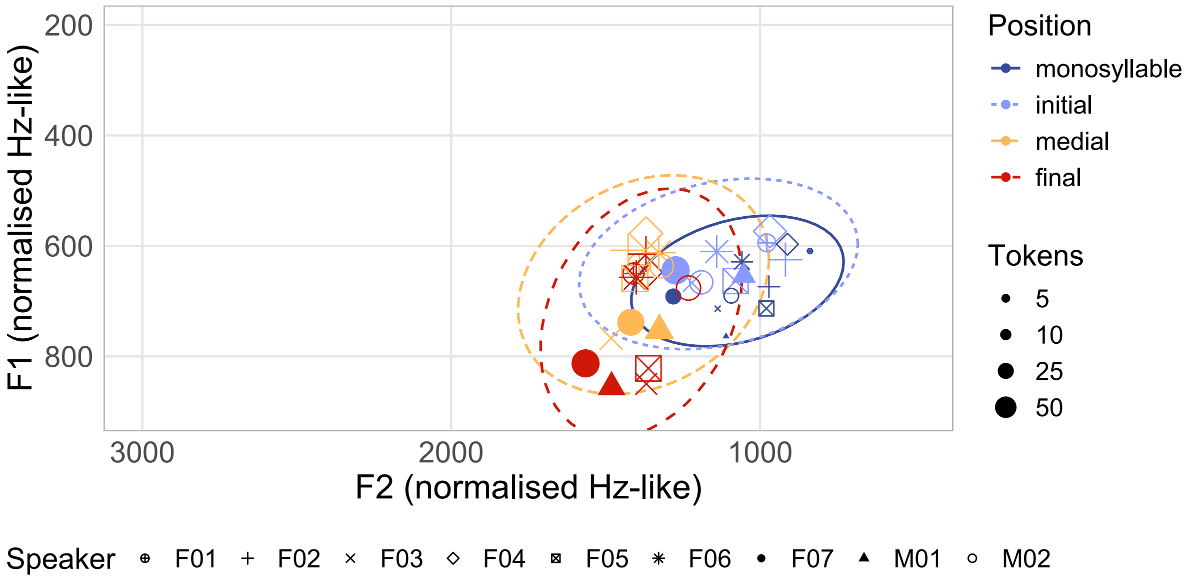


Figure 30: $F1 \times F2$ space for /ɑ/ ($N = 1846$) by position monosyllable/initial/medial/final. The unrounded allophone [ɑ] appears in non-initial syllables; the rounded [ɒ] is restricted to initial syllables. Points show aggregated medians for each speaker, and ellipses show the spread of the data (75% confidence).

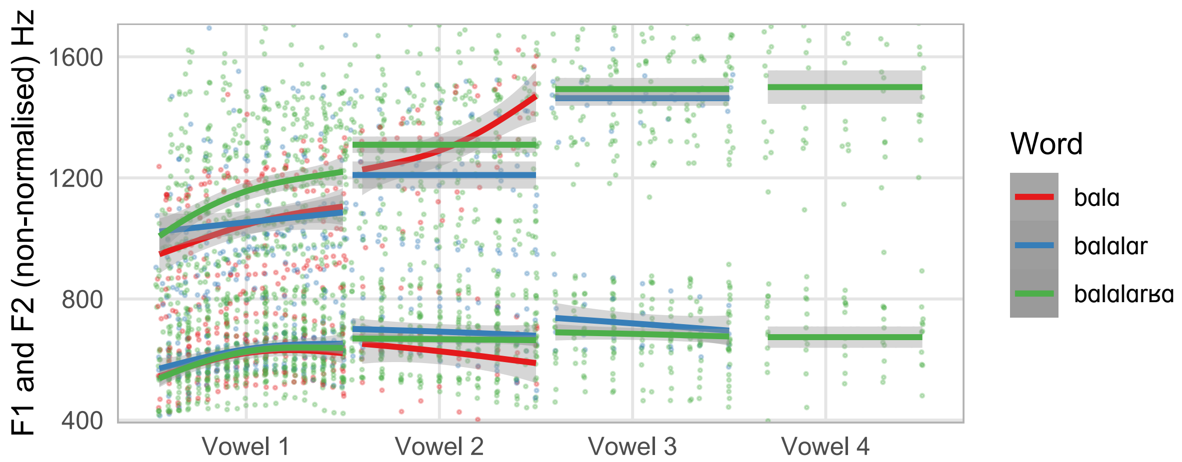


Figure 31: $F1$ and $F2$, in Hz, measured in Praat (automatically, with manual adjustment of formant ceilings) for /ɑ/ ($N = 73$) by position in 3 unique words, non-normalised, over 7 female speakers. The x-axis groups measurements by vowel position, and normalises durations of individual vowels; raw data is shown together with GAM-smoothed trajectories.

[18] F3 was judged not to be reliable as a cue to rounding in this test, both due to the presence of the velarised lateral and the uvular fricative, and due to the concerns raised above for the rounded vowels.

Mid vowels

There are four vowels which we categorise as neither low nor high: front unround /e/, front rounded /ø̞/, central /ə/, and back rounded /o/. /e/ is typically ⟨e⟩ throughout the descriptive literature, although Şahan Güney (2015: 22) uses ⟨é⟩, and Berta (1998: 283) treats this vowel as systematically reduced, and uses ⟨ĕ⟩. /ə/ is often ⟨ï⟩ (Poppe 1968: 9, Berta 2022: 303); Berta (1998: 283) again marks reduction across the board and uses ⟨ï̆⟩, Şahan Güney (2015: 22) uses ⟨ı⟩, and Comrie (1997: 900) uses ⟨ə⟩ as we do here. /ø̞/ and /o/ are commonly ⟨ö⟩ and ⟨o⟩ respectively, or reduced ⟨ö̆⟩ and ⟨ŏ⟩ for Berta (1998: 283).

The mid vowels share several properties and a diachronic origin, deriving from historical /i/, /y/, /ɨ/, and /u/ respectively as part of the Volga Turkic 'Great Vowel Shift', which also raised the historic mid vowels /e/, /o/, and /ø/ to /i/, /u/, and /ʉ/ respectively (for further details, see Berta 1989). Perhaps as a result, the mid vowels pattern unexpectedly like high vowels in other languages (including the other Turkic languages, save Bashkir) in at least two respects: in susceptibility to reduction and deletion, and in the action of vowel harmony, to which we return in a later section.

The mid vowels are subject to considerable reduction in unstressed syllables, and may delete entirely, especially adjacent to sibilants, and especially in connected speech: most speakers had ⟨кешенең⟩ /keʃeneŋ/ [k̩.ʃe.ˈneŋ] 'person-GEN' in the passage (in the sentence ⟨бу юлда яткан кешенең янында [...]⟩), but [ke.ʃe.ˈneŋ] in isolation. Deletions and devoicings are reported for the high vowels (that is, those corresponding to the mid vowels in Tatar[19]) in related languages (especially for Uigur, Hahn & Ibrahim 1991; Fiddler 2019; Mayer et al. 2022); further afield, such phenomena are cross-linguistically associated both with the high vowels and with durational effects (Gordon 1998). Previous descriptions invariably mention that the mid vowels in Tatar are distinctively shorter than the high vowels, and sometimes also refer to them as low-intensity, and either centralised or laxed (Comrie 1997: 900; see Baĭchura 1960: 41–42, 49–50 for measurements of duration, largely in agreement with ours below). /ə/ is the vowel most consistently described in the literature as short (Kurbatov et al. 1969: 85) or reduced (Poppe 1968: 9).

Due to effects of this type, Baskakov (1988: 136) argues that /ɨ i u ʉ/ and /ə e o ø̞/ are distinguished primarily by length, rather than by quality; counter to this, Berta (1998: 283) argues that durational differences are present but non-contrastive. Similar claims have been made for short high monophthongs in Kazakh (Washington 2016; McCollum 2015, 2018; for discussion, see McCollum & Chen 2020). Figure 32 shows the duration in milliseconds of all unstressed vowels separated by position, with a clear reduction in duration separating the mid vowels from both the low and high vowels in initial, medial, and final position in polysyllabic words (noting the absence of /ø̞/ and very low token numbers for the other round vowels in

[19] Fiddler (2019) suggests (see also Mayer et al. 2022) that the devoicing in Uigur is post-lexical phonology, phonologised relatively early in the history of Turkic. If this is the case, then one interpretation of the facts in Tatar is as a post-lexical rule targeting this class that predated the Volga vowel shift.

non-initial position), but not consistently in monosyllables.

In the initial syllable of a polysyllabic word, mean duration = 64 ms ($SD = 39$) for mid vowels vs 101 ms ($SD = 60$) for high vowels and 106 ms ($SD = 48$) for low vowels; in medial syllables, mean duration = 60 ms ($SD = 32$) for mid vowels vs 90 ms ($SD = 51$) for high vowels and 89 ms ($SD = 37$) for low vowels. These effects were statistically significant in both positions (initial: mid—high $t = -12.96$, $p < .001$, Cohen's $d = -0.743$; mid—low $t = -17.97$, $p < .001$, Cohen's $d = -0.956$; medial: mid—high $t = -7.89$, $p < .001$, Cohen's $d = -0.880$; mid—low $t = -20.71$, $p < .001$, Cohen's $d = -0.861$, with Bonferroni correction for multiple comparisons $\alpha = 0.05/8 = 0.00625$). In monosyllables, mid vowels (158 ms, $SD = 59$) showed less consistent reduction relative to high (178 ms, $SD = 89$) and low vowels (206 ms, $SD = 81$), and statistical significance was not achieved for the mid/high comparison (mid—high $t = -1.86$, $p = .259$, Cohen's $d = -0.249$; mid—low $t = -5.22$, $p < .001$, Cohen's $d = -0.633$, both after Bonferroni correction); /ə/ alone is distinctively short in monosyllables, to the exclusion of the mid vowels as a category. We suggest tentatively therefore that an analysis of these facts as deriving from the interaction of a height contrast and a process of gradient phonetic reduction outside stressed syllables is a better fit than an analysis as a true duration contrast.

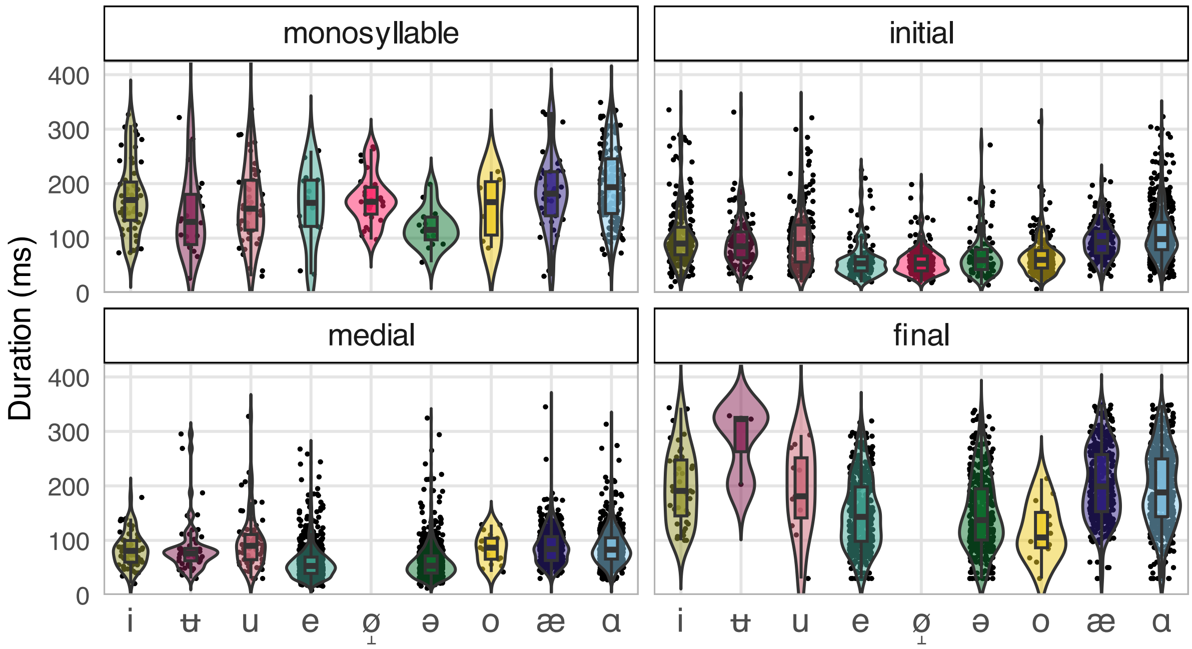


Figure 32: Duration in milliseconds for $N = 6950$ vowels across all speakers, by position within the word; the distribution of tokens is shown, along with the median (central black line) and the 25–75 inter-quartile range. 173 points were excluded due to unrealistically long durations (> 350ms) caused by hesitation, or due to artificiality of speech (e.g. syllabification tasks).

Claims made in the literature that these vowels are notably low-intensity in initial syllables, e.g. 'pronounced with a very weak spread of acoustic energy' (Johanson 2021: 227) were not supported; mean intensity showed mixed patterns across positions, with mid vowels not significantly distinguishable from high vowels ($t = 2.35$, $p = .163$, 1.39 dB difference; all comparisons with Bonferroni correction $\alpha = 0.05/8 = 0.00625$) and very slightly more intense than low vowels ($t = 4.59$, $p < .001$, 2.50 dB difference) in monosyllables, and more intense than high vowels ($t = 5.42$, $p < .001$, 1.75 dB difference) in initial position but not significantly different from low vowels ($t = 1.57$, $p = .930$, 0.49 dB difference).

High vowels

There are three high vowels, front unrounded /i/, central rounded /ʉ/, and back rounded /u/. /i/ is generally ⟨i⟩ in the descriptive literature, and /ʉ/ and /u/ generally ⟨ü⟩ and ⟨u⟩. As described above for the mid vowels, these high vowels derive from the raising of historic /e/, /ø/, and /o/ respectively during the Volga 'Great Vowel Shift'. Central rounded /ʉ/ is front for the purpose of all phonological alternations that reference vowel place, but is clearly central (Figure 25).

Vowel harmony

Like virtually all the Turkic languages, Tatar has a robust system of backness (or palatal) harmony (Poppe 1968: 15–16; Kurbatov et al. 1969: 96–97; Zakiev 1993: 170–173; Şahan Güney 2015: 32–33; Burbiel 2018: 10), which is consistently marked in the orthography. The existence of rounding (or labial) harmony in the language is disputed, and is discussed further below. Backness harmony applies both as a distributional restriction within roots and across morphological boundaries: roots typically contain only vowels from either the phonologically-front set of vowels, /i ʉ e ø̞ æ/, or the phonologically-back set, /u ə o ɑ/, illustrated in Table 5, but there is a significant proportion of disharmonic borrowed roots, e.g. ⟨гөнаh⟩ [gø̞.ˈnɑh] 'sin', ⟨цитата⟩ [(t͡)si.t̪ɑ.ˈt̪ɑ] 'citation'. Despite the very similar positions of /ə/ and /ʉ/ in phonetic space, they pattern differently: /ə/ is consistently back for the purposes of all vowel-related phonology (including the distribution of consonantal allophones, as discussed above), and /ʉ/ consistently front in the same respects.

Across morphological boundaries, the general pattern is that all affix vowels must agree for backness with the *final* vowel in the root, with the latter restriction evident from disharmonic cases, ⟨кадәр-ле⟩ /kɑd̪ær-lE/ [qɒ.d̪ær.ˈle] 'degree-ADJZ' *[qɒ.d̪ær.ˈlə] (on disharmony, see Poppe 1968: 16). Suffixes are thus underspecified for backness and typically belong to one of three classes: those that alternate [æ–ɑ], often written by Turcologists in underlying forms with A, to indicate the lack of underlying [±back] specification; those that alternate [e–ə] (likewise E), and those that alternate [ʉ–u] (likewise U), with the last being essentially restricted to the verbal noun suffix /-Uw/, /bɑr-Uw-E/ [bɒ.ru.ˈwə] ⟨бар-у-ы⟩ 'go-VN-POSS.3S' vs. /iz-Uw-Em-nAn/ [i.zʉ.wem.ˈnæn] ⟨из-ү-ем-нән⟩ 'press-VN-POSS.1S-ABL'). The typical pattern is illustrated in Table 6. A few borrowings trigger violations, however, as in /ʑæmgəjɑt̪-kE/ [ʑæm.ʁə.jɑt̪ʲ.ˈkæ] ⟨җәмгыять-кә⟩ 'community-DAT' from Arabic, which exceptionally takes the back suffix rather than the front; these exceptions are typically orthographically marked with the Cyrillic ‘soft sign’, ⟨ь⟩.

Table 5: Backness harmony within roots.

First vowel	Following vowel	Orthographic	Example word	Gloss
æ	æ	чәчәк	[ɕæ.ˈɕæk]	flower
	e	хәреф	[χæ.ˈref]	letter
e	æ	белән	[be.ˈlæn]	with (*passage*)
	e	кеше	[ke.ˈʃe]	person
i	æ	кинәт	[ki.ˈnæt̪]	suddenly (*passage*)
	e	сигез	[si.ˈgez]	eight
ɑ	ɑ	агач	[ɒ.ˈʁɒɕ]	tree
	ə	адым	[ɒ.ˈd̪əm]	step
ə	ɑ	пычак	[pə.ˈɕɑq]	knife
	ə	кызыллык	[qə.zəɫ.ˈɫəq]	blush

Table 6: Backness harmony across morpheme boundaries.

Root vowel	Suffix vowel	Orthographic	Example word	Gloss
æ	æ	(кеше) әйт-кән	[(k̹ʃe) æjt̪.ˈkæn]	(person) say-PST.3s
	e	чәчәк-не	[ɕæ.ɕæk.ˈne]	flower-ACC
e	æ	кер-гән	[ker.ˈgæn]	enter-PST.3s
	e	сез-ңе	[sez.ˈneŋ]	you-GEN
i	æ	ди-гән	[d̪i.ˈgæn]	say-PST.3s
	e	дин-е	[d̪i.ˈne]	religion-POSS.3s
ɑ	ɑ	ят-кан	[jɒt̪.ˈqɒn]	lie-PST.3s
	ə	хат-ны	[χɒt̪.ˈnə]	letter-ACC
ə	ɑ	чык-кан	[ɕəq.ˈqɑn]	go.out-PST.3s
	ə	кыз-ы	[qə.ˈzə]	daughter-POSS.3s (*in carrier sentence*)

Labial/rounding harmony

Like backness harmony, rounding harmony is attested across the Turkic family, but is often (see, for example, McCollum & Kavitskaya 2022 on Crimean Tatar, and McCollum 2015; McCollum & Chen 2020 on Kazakh) subject to greater restrictions than backness harmony, and more likely to (have) be(en) lost in diachronic change. In Tatar, the previous descriptive literature often, though not always[20], claims that the mid round vowels /o/ and /ø̞/ trigger rounding in the mid unround vowels /e/ and /ə/ (Faseev 1966: 815, Kurbatov et al. 1969: 100; Berta 1998: 284, Zakiev 1993: 121, Şahan Güney 2015: 33–34, Burbiel 2018: 10); several authors include caveats with respect to apparent gradience, as e.g. Comrie (1997: 903), who refers to the degree of rounding 'decreas[ing] progressively through the word', and the literature is rather mixed in attributing this to phonetic, phonological, or phonotactic causes. (The most phonetically-detailed previous work on the language, Conklin & Dmitrieva (2019), refers to the 'unusual behavior' of sequences of mid vowels led by /o/ and /ø̞/, and as such excludes these from analysis.) No such pattern is marked in the orthography. Examples are listed in Table 7, for /ə/ and /e/ following round and unround vowels both root-internally and in affixes in disyllables.

Table 7: Rounding 'harmony' in roots and across morpheme boundaries.

Root vowel	Following vowel	Orthographic	Example word	Gloss
o	ə	борын кош-ы	[bo.ˈron] [qo.ˈʃo]	nose bird-POSS.3s
ø̞	e	дөрес көн-ең-не	[d̪ø̞.ˈrø̞s] [kø̞.nø̞ŋ.ˈnø̞]	correct day-POSS.2s-ACC
ə	ə	шыпырт кыз-ы	[ʃə.ˈpərt̪] [qə.ˈzə]	quietly (*passage*) daughter-POSS.3s
e	e	кеше сез-нең	[ke.ˈʃe] [sez.ˈneŋ]	person you-GEN
u	ə	туры дус-ы	[t̪u.ˈrə] [d̪u.ˈsə]	straight friend-POSS.3s
ʉ	e	түгел сүз-ен-дә	[t̪ʉ.ˈgel] [sʉ.zen.ˈdæ]	not word-POSS.2s-LOC

Figure 33 shows a subset of the data filtered to words with either a non-round (initial /i/, /e/, /ə/, and /æ/) or mid round (/ø̞/ and /o/) vowel in the initial syllable, and to tokens of the mid unround vowels /e/ and /ə/; although there is very considerable overlap in the realisations of these tokens, mid vowels preceded by round vowels earlier in the stem are shifted in $F1 \times F2$ space in a direction consistent with the relative relationships of the round/unround pairs in syllable 1. The relatively high place of /ø̞/ is a challenge for the acoustic detection of labial harmony: even a large effect of lip-rounding on the F2 of the vowel may well be masked by the effect of the vowel height target in a true /ø̞/ token. Apparent *raising* may therefore be a potential indicator of rounding in the /e/–/ø̞/ harmonic pair; at the same time, vowels may raise in unstressed syllables for reasons unrelated to either harmony or V-to-V coarticulation, or indeed, due to V-to-V coarticulation with the high vowels, and as such, we consider the description here (which does not fully account for distributional and coarticulatory confounds of this type) preliminary.

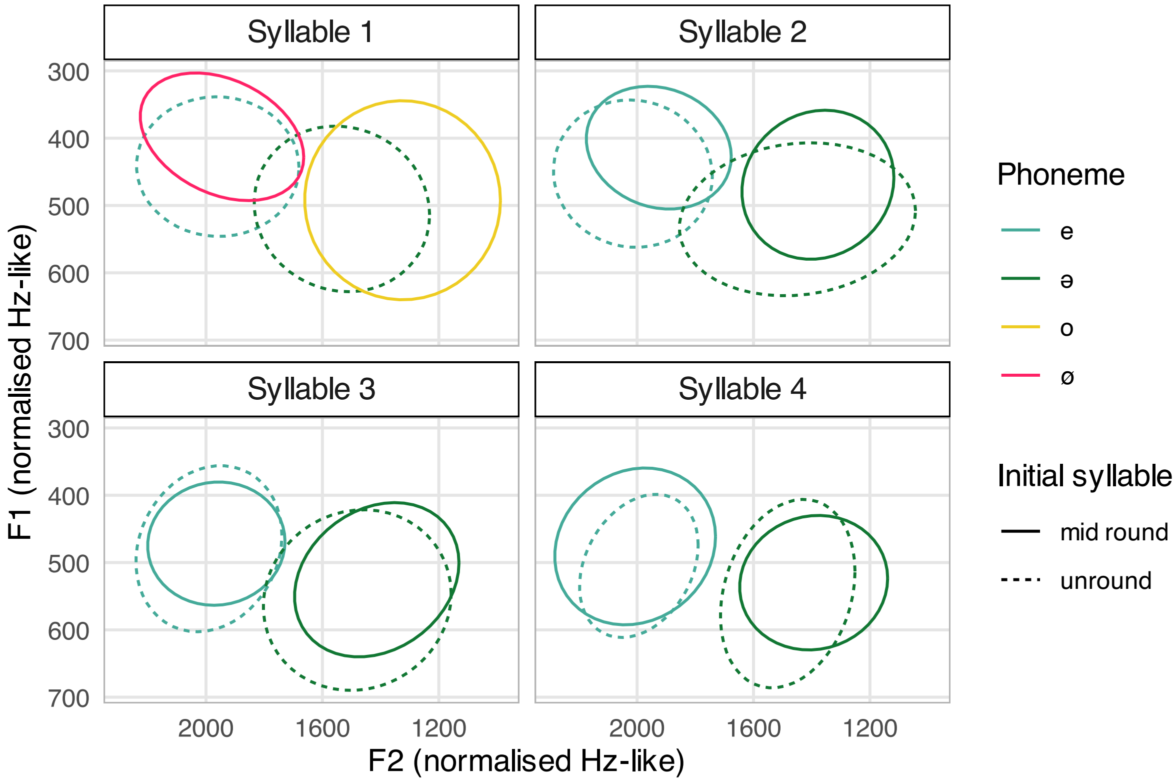


Figure 33: $F1 \times F2$ space for the mid vowels, /o/, /ø/, /e/, and /ə/, which are the predicted triggers and targets of rounding harmony, showing 75% confidence ellipses. ($N = 2475$)

We categorised our data by the content of the initial 'trigger' syllable: non-round (initial /i/, /e/, /ə/, and /æ/), versus mid round (/ø/ and /o/), versus high round (/u/ and /ʉ/), versus low round (the invariably phonetically rounded first-syllable /ɑ/), and fit mixed-effects linear regression models to F1, F2, and F3 with `lme4` and the Satterthwaite approximation implemented in `lmerTest`; models included fixed effects for trigger category, vowel position (syllable number 2, 3, 4, treated as continuous), phoneme identity (/e/ vs. /ə/), and their interactions, with stress as a control, plus random intercepts for speaker, preceding consonantal context, and following consonantal context ($N = 2{,}924$ observations over 1611 words; only 113 of 3,037 relevant tokens were in syllables 5 and 6, and these syllables were thus excluded due to skewed distribution). These results are visualised in Figure 34.

Mid round triggers lowered $F1$ in /e/ by approximately 28 Hz ($\beta = -42.07 + 14.50 = -27.57$, main effect $\beta = -42.07$, SE = 9.36, $t(2834) = -4.493$, $p < .001$) and high round triggers by

[20] The sole truly countervailing position is taken by Conklin (2015a: 53–62), who finds neither categorical rounding harmony nor gradient coarticulation in data from the front vowels for a single speaker. Conklin's measures are precisely the inverse of ours: that is, she takes F2 as the best measure of rounding within the front vowels, and F1 within the back vowels.

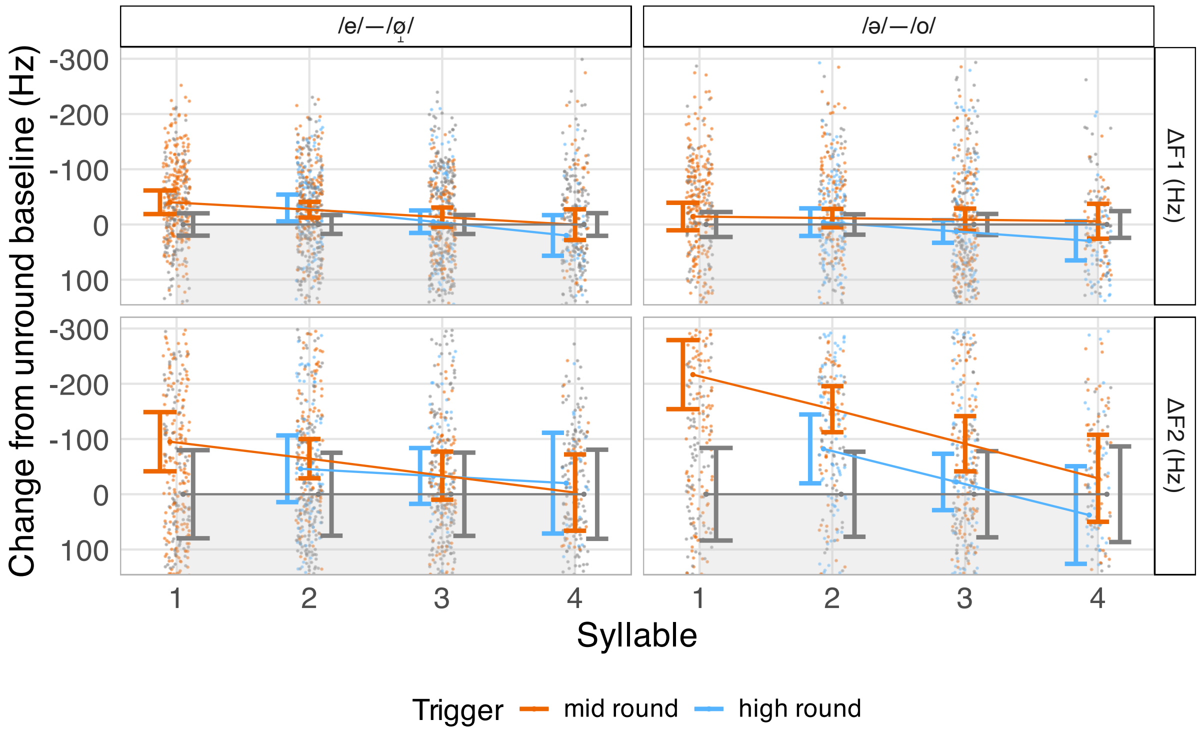


Figure 34: Predicted change $\Delta F1$ (top) and $\Delta F2$ (bottom) for target mid vowels with mid round triggers (/ø̞/, /o/; orange) or high round triggers (/u/, /ʉ/; blue), relative to unround baseline (/i/, /e/, /ə/, /æ/; grey). The contrast shown in syllable 1 is between orthographic and phonemic /e/, /ə/ (grey) and orthographic/phonemic /ø̞/, /o/ (orange); in other syllables, all tokens are of *orthographic* /e/, /ə/. Lines show model predictions from mixed-effects regression with 95% confidence intervals; small background points show raw observations, jittered horizontally for visibility.

31 Hz ($\beta = -57.01 + 25.87 = -31.14$, main effect $\beta = -57.01$, SE = 19.03, $t(2890) = -2.995$, $p = 0.003$) in syllable 2; the effect of mid round triggers decreased by 14 Hz per syllable ($\beta = 14.50$, SE = 5.83, $t(2880) = 2.489$, $p = 0.013$) and high round triggers by 26 Hz per syllable ($\beta = 25.87$, SE = 10.37, $t(2892) = 2.495$, $p = 0.013$). Low round triggers had no effect on /e/, and no $F1$ effects reached significance for /ə/. Mid round triggers lowered $F2$ in /e/ by 66 Hz ($\beta = -95.66 + 29.55 = -66.11$, main effect $\beta = -95.66$, SE = 22.73, $t(2875) = -4.208$, $p < .001$) in syllable 2, with a decrease in effect of 30 Hz per syllable ($\beta = 29.55$, SE = 14.09, $t(2892) = 2.097$, $p = 0.036$). Mid round triggers lowered $F2$ in /ə/ by approximately 174 Hz ($\beta = -107.96$, SE = 35.04, $t(2896) = -3.081$, $p = 0.002$ for the interaction) in syllable 2. High round and low round effects did not reach significance.

We claim therefore that there is indeed a statistically significant effect of initial-syllable rounded mid vowels on the following mid vowels, especially back ones. Further work will need to determine whether or not the pattern is truly iterative over a larger domain and to disambiguate between the competing hypotheses of phonologically-controlled harmony and phonetically-controlled coarticulation, however, and to better understand the state of sequences in which mid vowels follow *high* round triggers. True harmony with these parameters is at least somewhat typologically unusual (Kaun 2004): while non-high triggers and trigger-target matches favour rounding harmony cross-linguistically, non-high *targets* disfavour it (in Kaun's framing, perceptually-weak vowels are good triggers, but bad targets; though see McCollum 2018 for complications).

Syllable structure

Most Tatar monosyllables are VC or CVC; while not altogether unattested, there are very few monosyllabic words consisting solely of an onset and a nucleus CV, or of a nucleus alone V. In polysyllabic words, however, CV syllables predominate, and CVC is also frequent. Onsetless V and VC syllables are possible, but less frequent than those with onsets, and are not well-tolerated word-initially: word-initial vowels, even epenthetic ones, are often repaired with an initial glottal stop. (C)(C)V(C)(C) thus gives the maximal set of all possible syllables (Faseev 1966: 817–818; Zakiev 1993: 85–87; Burbiel 2018: 16): Zakiev (1993: 86) gives an indicative distribution of types, V 5.1%, VC 4.2%, CV 46.5%, CVC 43.2%, VCC 0.3%, CVCC 0.4%, CCV 0.2%, CCVC 0.1%[21]. As this last suggests, complex codas are possible, but are highly restricted, and syllables containing both a complex onset and a complex coda are essentially absent. As in Kazakh (McCollum & Chen 2020: 16), the first consonant must be a sonorant, and the second must be voiceless, as in ⟨карт⟩ [qɒrt̪] 'old'; epenthetic repairs are invariably applied to loans that violate this condition, as ⟨жанр⟩ [ˈʒɒnər] 'genre'. Complex onsets are only attested in loans, like ⟨клуб⟩ [ˈqɫub] 'club', and are often realised with an epenthetic vowel.

[21] For a discussion of syllable structure in Tatar onomatopoeia specifically, see Károly (2024: 639–640).

Table 8: Syllable types in our data

Syllable type	Orthographic	Word	Gloss
V	ә	[ˈæ]	but; yet
	ата	[ɒ.ˈt̪ɒ]	father
VC	ат	[ˈɒt̪]	horse; name
	өч	[ˈø̞ɕ]	three
CV	бу	[ˈbu]	this
	дала	[d̪ɒ.ˈɫɒ]	flat land
CVC	күк	[ˈkʉk]	sky; blue
	тыш	[ˈt̪əʃ]	outside
VCC	ант	[ˈɒnt̪]	oath
	дүртенче	[d̪ʉr.t̪en.ˈɕe]	fourth
CVCC	шыпырт	[ʃə.ˈpərt̪]	quietly
	карт	[qɒrt̪]	old

Stress & intonation

Stress

The descriptive literature has generally claimed that stress in Tatar is word-final, with stress shift onto suffixes, but subject to several exceptions (Faseev 1966: 818–819; Poppe 1968: 14–15; Kurbatov et al. 1969: 99; Zakiev 1993: 96–102; Şahan Güney 2015: 35–39; Burbiel 2018: 17–21). Word-level final stress has typically been assumed in acoustic work on the language, e.g. in the description of ⟨ый⟩ given by Conklin & Dmitrieva (2019). Figure 26 suggests that vowel qualities are more peripheral in final syllables than in initial or medial syllables across the board, and Figure 32 shows that vowels in monosyllables and final syllables are distinctively longer than those in initial and medial syllables of polysyllabic words. It is widely agreed in the descriptive literature that certain suffixes and postpositions do not bear word or phrase-level stress: see, for example, ⟨кеше белән⟩ [kĕ.ˈʃe‿bĕ.læn] *[kĕ.ˌʃe‿be.ˈlæn] 'person with.PP' in the recorded passage. The negative suffix /-mA/ has been claimed to block stress placement and the rightward shift of stress: ⟨чыкма⟩ [ˈɕəq.mɑ] 'go.out-NEG.IMP.2S' in the recorded passage, and ⟨бармадыгыз⟩ [ˈbɒr.mɒ.d̪ə.ʁəz] 'go-NEG-PST.2P'. In most compounds, the second constituent carries the main stress, as e.g. ⟨сыр заводы⟩ [ˌsər‿zɒ.vo.ˈd̪ə] 'cheese factory-POSS.3S'. Primary stress in coordinating compounds, however, falls on the final syllable of the first constituent: ⟨апалы-энеле⟩ [ɒ.pɒ.ˈɫə.e.ne.ˌle] 'older sister with younger brother'. It is unclear, however, how these judgments have been made in the past, and how the determination of word-level stress interacts with phrase-level intonation (which see Royer 2017; Royer & Jun 2018, 2019).

Table 10: Example test items for stress.

Orthographic	Transcription excluding stress	Gloss
тел	/t̪el/	tongue
телләр	/t̪el-lær/	tongue-PL
телләре	/t̪el-lær-e/	tongue-PL-POSS.3S
телләрендә	/t̪el-lær-en-d̪æ/	tongue-PL-POSS.3S-LOC
телләрендәге	/t̪el-lær-en-d̪æ-ɡe/	tongue-PL-POSS.3S-LOC-REL
телләрендәгечә	/t̪el-lær-en-d̪æ-ɡe-ɕæ/	tongue-PL-POSS.3S-LOC-REL-EQ

To better characterise non-exceptional stress patterns, we tested 43 words per speaker in the carrier sentence ⟨Ул [X] сүзен кабатлады⟩ /ul [X] sʉzen qɑbɑt̪lɑd̪ə/: eight roots (⟨тел⟩ 'tongue; language', ⟨дин⟩ 'religion', ⟨кыз⟩ 'girl', ⟨кош⟩ 'bird', ⟨көн⟩ 'day', ⟨дус⟩ 'friend', ⟨сүз⟩ 'word', ⟨бала⟩ 'child') were used with between zero and five suffixes, creating test items with a maximum of 6 syllables, of which examples are given in Table 10. For this purpose, we disregarded the remainder of our data, which was elicited in isolation or otherwise not controlled for possible interference from focus marking and other intonational phenomena. 44 words were tested originally, but one test item with more than 6 syllables was excluded from the results as all speakers hesitated on it (⟨телләрендәгечәлегенә⟩ /t̪ellærend̪æɡeɕælegenæ/ 'tongue-PL-POSS.3S-LOC-REL-EQ-DER-POSS.3S-DAT'). Of the 6 speakers who carried out this task, 2 speakers (M02 and F06) were excluded, due to clear hesitations and prosodic breaks within the carrier sentences that meant that we could not satisfactorily rule out the effects of unpredictable phrase-level intonational phonology on the final measures. Maximum $f0$ (in Hertz), mean intensity over the entire duration of the vowel (in decibels), and duration were measured over $N = 488$ target vowels.

Figure 35 shows box-plots for duration, maximum $f0$ (z-scored), and mean intensity, aggregated across speakers and shown by position initial/medial/final; monosyllables are coded as final for this purpose (but clearly separable by word length). There is a clear rise in duration from the initial to the final syllable, which is suggestive; final-vowel lengthening, however, is cross-linguistically frequent and not necessarily inevitably correlated with stress (White et al. 2020). The situation of maximum $f0$ is inconclusive on the basis of this plot, and there is a slight decrease in intensity from the initial to the final syllable.

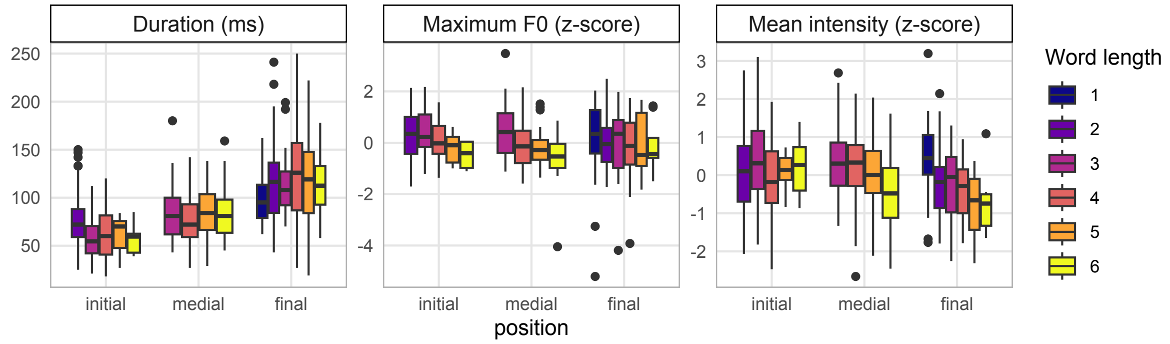


Figure 35: Duration (in milliseconds), maximum *f*0 (z-scored speaker-intrinsically), and maximum intensity (z-scored) across ternary positions (word-initial, word-medial, word-final) for words of different lengths (1-6 syllables, indicated by colour).

We fit mixed-effects models to each variable (duration, maximum *f*0, and mean intensity), using syllable position, speaker, and word length as fixed effects, including all interactions, and random intercepts for lexical root. The significance of polynomial terms and the selection of effects was determined by likelihood ratio testing; quadratic models were selected for all dependent variables (duration: $\chi^2(8) = 21.699$, $p < 0.01$ vs. linear; *f*0: $\chi^2(8) = 33.1635$, $p \ll 0.001$ vs. linear; intensity: $\chi^2(8) = 26.396$, $p < 0.001$ vs. linear). Figures 36 to 38 show predictions, and disaggregate the data by speaker and by individual syllable number, within words of all lengths. Figure 36 straightforwardly confirms the final-vowel lengthening seen in Figure 35; for all speakers, duration increased across the word (F02: 76.5 ms, $p < .001$; F04: 48.2 ms, $p < .001$; F05: 22.5 ms, $p = 0.037$; F07: 29.4 ms, $p = 0.006$), with the greatest effect in shorter words.

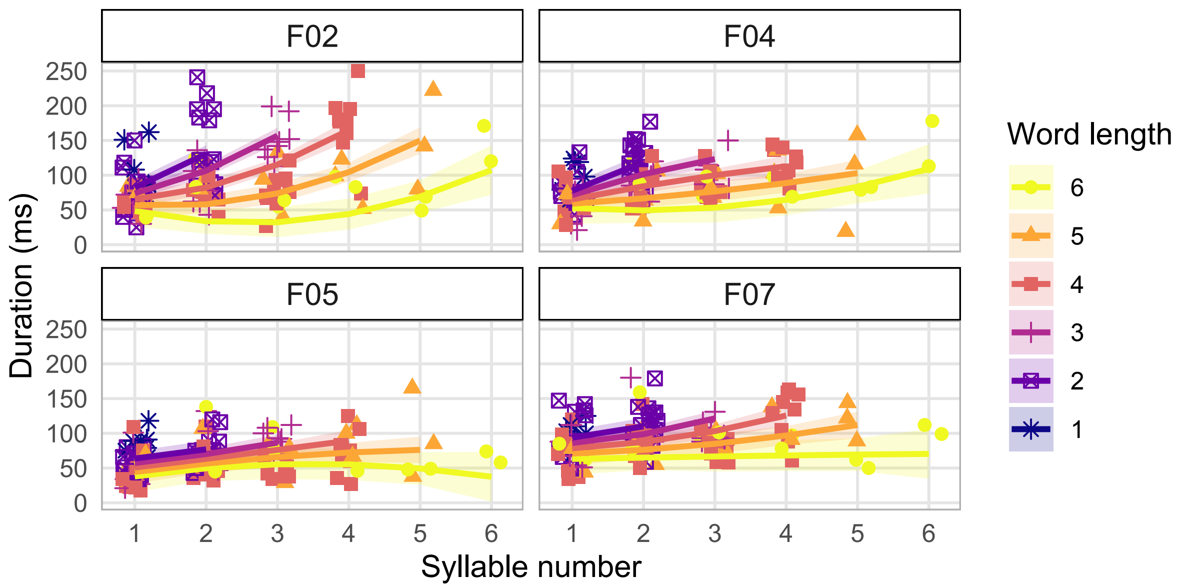


Figure 36: Duration (in milliseconds), separated by speaker, for 488 vowels in 43×4 words of different lengths; points represent individual measurements, and trend-lines represent the predictions of the mixed-effect models described above, for each length of word (1–6 syllables).

Figure 37 suggests that the inconclusive patterns in maximum *f*0 shown above result, in fact, from a superposition of several speaker-specific patterns. For speaker F05, there is a rise towards the final syllable that suggests the presence of a high pitch target at the right edge (final vs. initial +0.655 z, $p = 0.053$). For F02, there was no statistically-significant difference between initial and penultimate syllables, but a sharp decrease in *f*0 (final vs. initial −0.940 z, $p = 0.005$) in the final syllable, consistent with the realisation of a *low* pitch target. F07 showed declining F0 throughout the word, with both penultimate (initial vs. penultimate −0.584, $p = 0.053$) and final syllables (initial vs. final −0.620, $p = 0.066$) lower than initial syllables. F04 showed no significant positional F0 differences. Royer (2017); Royer & Jun (2018, 2019) describe (presumed) final stressed syllables in Tatar as marked by post-lexical [L+H*] pitch accents[22], with the F0 peak 'often' (but thus presumably not always) aligned with the end of the final syllable, and a low target in the penultimate syllable; this describes F05 and perhaps F04, but not all our speakers.

Figure 38, again, shows several speaker-specific patterns in intensity. F02 showed significantly lower intensity on final syllables compared to initial syllables (final vs. initial -1.16 z, $p < .001$s), but penultimate syllables were not statistically distinguishable from initial syllables. F07 showed a similar but only marginally significant pattern (final vs. initial -0.589, $p = 0.082$; no significance penultimate vs. initial). For F05, penultimate syllables showed marginally lower intensity than initial syllables (-0.536, $p = 0.076$) but final syllables were not distinguishable from initial syllables, suggesting a high pitch target at the right edge. F04 showed no significant positional intensity differences.

[22] Royer & Jun (2018: 771) further describe a pattern in which high F0 and high intensity appear on the initial syllable when the word is the last or second-to-last in an intonational phrase and preceded by a pitch-accented word; this description does not seem to apply to our test set.

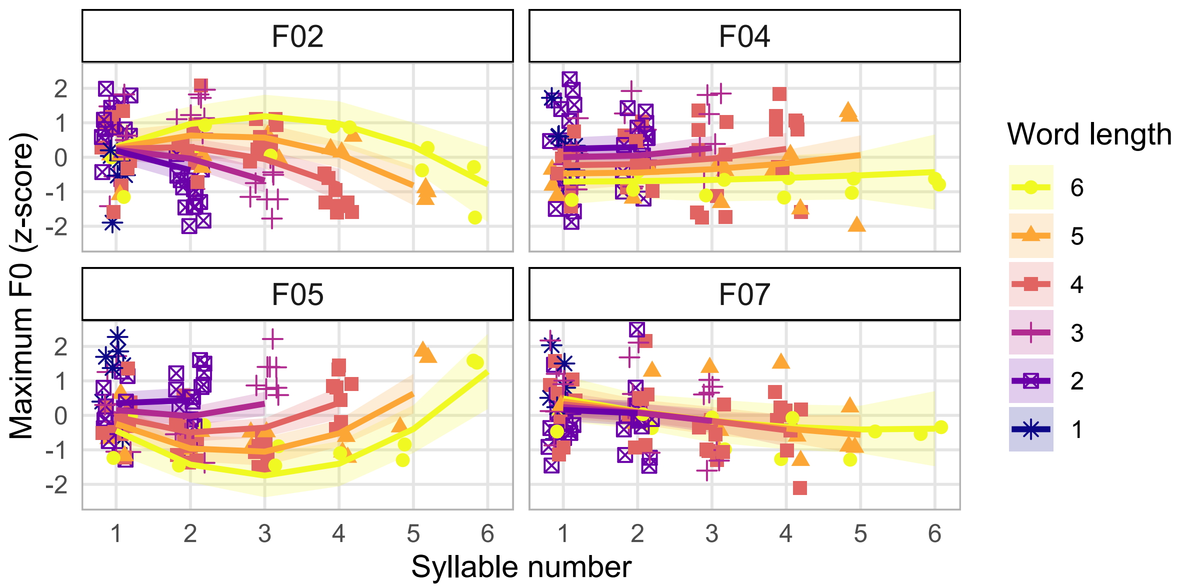


Figure 37: *f*0 (z-scored), separated by speaker, for 488 vowels in 43 × 4 words of different lengths; points represent individual measurements, and trend-lines represent the predictions of the mixed-effect models described above, for each length of word (1–6 syllables).

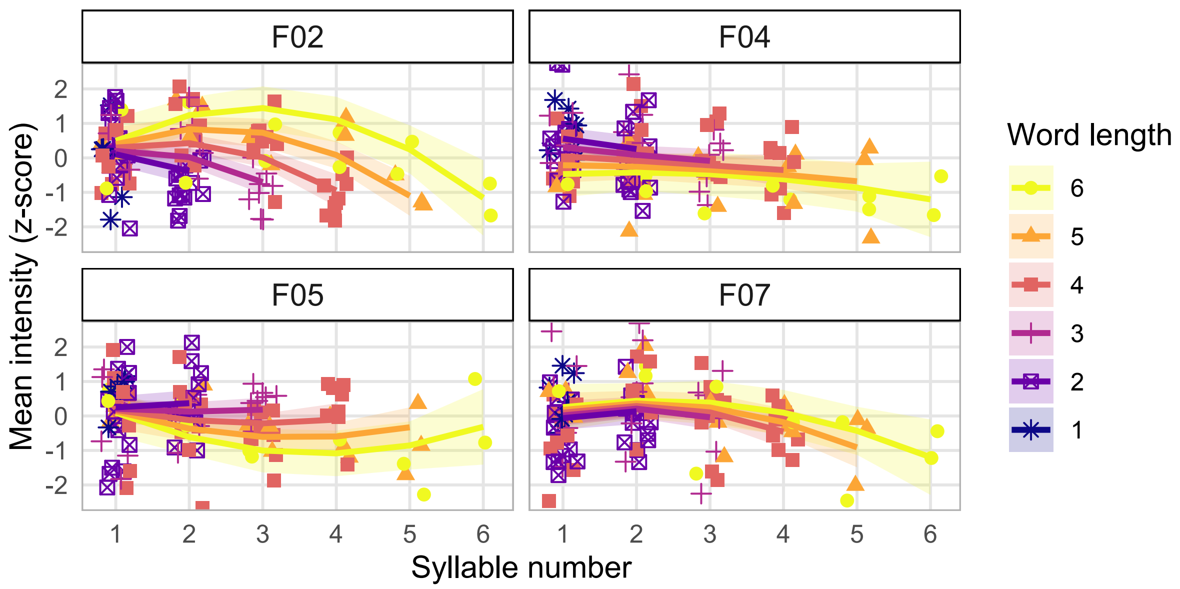


Figure 38: Intensity (z-scored), separated by speaker, for 488 vowels in 43 × 4 words of different lengths; points represent individual measurements, and trend-lines represent the predictions of the mixed-effect models described above, for each length of word (1–6 syllables).

These results are not *inconsistent* with the diagnosis of final stress, but it is clear that they present a mixed picture. In particular, individual participants used the acoustic cues that we investigate differently. Our small sample does not allow us to comment further on whether these differences are explained by either intonational phonology or appeal to sociolinguistic stratification, but see Borise & Georgieva (2024) for a discussion of a similar diversity of acoustic cues marking stress in Udmurt (a language unrelated to Tatar but geographically and typologically close to it), which is also typically considered to have final stress. Ido (2025) reports long final vowels, penultimate high *f*0 followed by low final *f*0, and low final intensity for a speaker of the Turkic language Uzbek, which, like Tatar, has typically been analysed as a final-stress language with a few exceptional stress-repelling morphemes. For Kazakh, McCollum & Chen (2020) report a rather unclear situation (summarising a complex picture, final syllables have low *f*0, low intensity, and long duration; initial syllables have high intensity, high *f*0, and short duration) for their reference speaker, and as such refrain from diagnosing stress of any kind in the language; the descriptive literature on Kazakh overall has claimed either final stress, word-final pitch accent combined with word-initial stress accent (Kirchner 1998), or no word-level stress but phrase-level prominence only (Vajda 1994: 644–646), and taken together, these accounts may well represent similar speaker-level variation in the realisation of phonological stress to that which we report here for Tatar.

Intonation

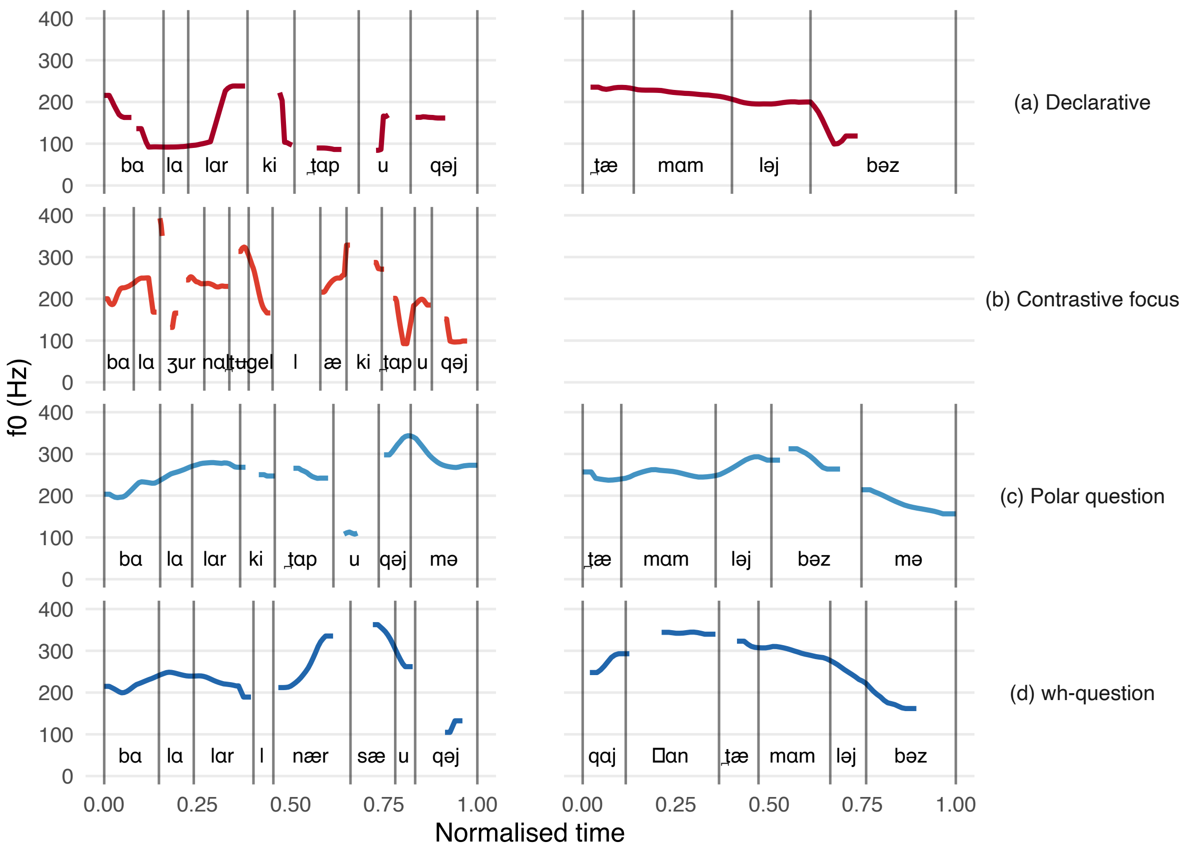


Figure 39: Smoothed F0 contours for (a) declarative, (b) contrastive focus, (c) polar question, and (d) *wh*-question utterances produced by speaker F07. F0 measurements were taken at 10ms intervals and smoothed using a rolling mean in runs of > 5 consecutive points. Time is normalised from 0 to 1 for each utterance, and syllable boundaries marked with vertical lines.

Figure 39 shows *f*0 contours for (a) two declaratives, ⟨Балалар китап укый.⟩ /bɑlɑlɑr kit̪ɑp uqəj/ ‘The children are reading a book.’ (child-PL book read-PRS.3) and ⟨Тәмамлыйбыз⟩ /t̪æmɑmləjbəz/ ‘We are finishing.’ (finish-PRS-1p), (b) one contrastive focus sentence ⟨Бала журнал түгел, ә китап укый.⟩ /bɑlɑ ʒurnɑl t̪ʉgel, æ kit̪ɑp uqəj/ ‘The child is reading a book, not a magazine.’ (child magazine not but book read-PRS.3), (c) two polar questions, ⟨Балалар китап укыймы?⟩ /bɑlɑlɑr kit̪ɑp uqəjmə/ ‘Are the children reading a book?’ (child-PL book read-PRS.3-Q) and ⟨Тәмамлыйбызмы?⟩ /t̪æmɑmləjbəzmə/ ‘Are we finishing?’ (finish-PRS-1p-Q), and (d) two *wh*-questions, ⟨Балалар нәрсә укый?⟩ /bɑlɑlɑr nærsæ uqəj/ ‘What are the children reading?’ (child-PL what read-PRS.3) and ⟨Кайчан тәмамлыйбыз?⟩ /qɑjɕɑn t̪æmɑmləjbəz/ ‘When are we finishing?’ (when finish-PRS-1p).

Simple declaratives (a) show falling *f*0 across the sentence (described by Royer & Jun 2018 as a L% boundary tone); a right-edge (final-syllable) raised *f*0 can be seen on the subject /bɑlɑlɑr/ ‘child-PL’ (consistent with L+H*). Utterance-medial raised *f*0 is also associated with continuation, as in (b) /t̪ʉgel/ ‘not’, and with the presence of the question marker /-m(e/ə)/ and the *wh*-words /nærsæ/ ‘what’, /qɑjɕɑn/ ‘when’.

Transcription of recorded passage

The following transcriptions represent the short folk tale *Куркак юлдаш* ‘The cowardly companion’ (Zakirova 2011: 145), provided with this paper and produced by our main informant, F07. Both broad and narrow phonetic transcription are provided along with a version in the standard Tatar orthography, interlinear gloss, and English translation.

Phonemic transcription

elek:e zɑmɑndɑ ike keʃe | iptæʃ quʃələp | julʁɑ ɕəq:ɑn:ɑr ‖ bolɑrʁɑ zur | urmɑn ɑrqələ ʉtærgæ turə kilgæn ‖ urmɑnʁɑ kergæn:ær ikæn | bolɑrnəŋ qɑrʃəsənɑ kinæt kenæ ber ɑju kilep ɕəq:ɑn ‖ monə kʉrgæɕ bolɑrnəŋ berse | bik qurqəp ɑʁɑɕ bɑʃənɑ menep qɑɕqɑn ‖ ikenɕese | ʉlgæn buləp | tiz genæ suzələp jɑtqɑn ‖ ɑju | bu juldɑ jɑtqɑn keʃeneŋ jɑnəndɑ æjlænæ æjlænæ bɑʃlɑræn | ɑjɑqlɑrən isnægæn dæ dij | ber dæ timijɕæ ʉz julənɑ kitkæn dij ‖ ɑju kitkæn:æn soŋ | ælege keʃe ɑʁɑɕ bɑʃən:ɑn tø̞ʃep iptæʃen:æn sorɑʁɑn ‖ ɑju sineŋ qolɑʁənɑ ʃəpərt qənɑ nærsæ æjt:e | digæn ‖ bu keʃe æjtkæn ‖ ɑju miŋɑ | mon:ɑn soŋ ʃuʃəndəj qurqɑq kʃe belæn julʁɑ ɕəqmɑ | dip æjt:e digæn

Narrow transcription

elek:e zɑmɑndɑ‿i̯ke kə̆ʃe‿i̯ptæʃ | quʃəłəp | jułʁɑ ɕə̆q:ɑn:ɑr̥ ‖ bołɑrʁɑ‿zurə̆ | urə̆mɑn ɒɾqəłə‿wʉtærɣæ turə kilɣæn ‖ urə̆mɑŋɑ kerĕgæn:ær ikæn | bołɑrnəŋ qɒʃəsənɑ kinæt ke̝næ bə̆r ɒju kilĕp ɕəq:ɑn ‖ monə kʉrgæɕ bołɑrnəŋ ber̥se | bik qurqəp ʔɒʁɒɕ bɒʃnɑ menep qɒɕqɑn ‖ ikẽɕese̤ | ʉlgæm‿buləp | tiz genæ suzələp jɒtqɑn ‖ ɒju | bu jułdɑ jɒtqɑn kʃeneŋ jɒnəndɑ æjlænæ æjlænæ bɒʃlɑræn | ɒjɒqlɑrən isnægæn dæ dij | ber dæ timijɕæ ʉʒ‿juɫənɑ kitkæn dij || ɑju kitkæn:æ̃‿səŋ | æ̰lege kĕʃe ɒʁɒɕ b̥ɒʃn:ɑn tʃø̞p iptæʃen:æ̃‿soɾɑʁɑn || ɑju sineŋ qoɫɑɰəŋɑ ʃəpərt qənɑ nærsæ‿æ̆jt:e | digæn || bu kʃe æjtkæn || ɒju biŋɑ | mon:ɑn sõ‿ʃuʃundɨ quɾqɑq kʃe blæn juɫə̆ʁɑ‿ɕəqmɑ | diβ‿æjt:e diɣæn

Orthographic version

Элекке заманда ике кеше, иптәш кушылып, юлга чыкканнар. Боларга зур урман аркылы үтәргә туры килгән. Урманга кергәннәр икән, боларның каршысына кинәт кенә бер аю килеп чыккан. Моны күргәч, боларның берсе, бик куркып, агач башына менеп качкан. Икенчесе, үлгән булып, тиз генә сузылып яткан. Аю, бу юлда яткан кешенең янында әйләнә-әйләнә, башларын, аякларын иснәгән дә ди, бер дә тимичә үз юлына киткән ди. Аю киткәннән соң, элеге кеше агач башыннан төшеп иптәшеннән сораган: – Аю синең колагына шыпырт кына нәрсә әйтте? – дигән. Бу кеше әйткән: – Аю миңа, моннан соң шушындый куркак кеше белән юлга чыкма, дип әйтте, – дигән.

Interlinearised version

(1)	elek:e	zɑmɑndɑ	ike	keʃe	iptæʃ	quʃələp	julʁɑ	ɕəq:ɑn:ɑr
	former	time-LOC	two	person	friendship	be.united-VADV	road-DAT	go.out-PST-3p

Once upon a time, two men became friends, and went out together on a journey.

(2)	bolɑrʁɑ	zur	urmɑn	ɑrqələ	ʉtærgæ	turə	kilgæn
	they-DAT	big	forest	through.PP	pass-VADV	straight	come-PST.3s

They had to pass through a big forest.

(3)	urmɑnʁɑ	kergæn:ær	ikæn	bolɑrnəŋ	qɑrʃəsənɑ	kinæt	kenæ	ber	ɑju
	forest-DAT	enter-PST-3p	COP	they-GEN	against-POSS.3s-DAT	suddenly	just	one	bear
	kilep	ɕəq:ɑn							
	come-vadv	go.out.AUX-PST.3s							

When they entered the forest, a bear suddenly appeared before them.

(4)	monə	kʉrgæɕ	bolɑrnəŋ	berse	bik	qurqəp	ɑʁɑɕ	bɑʃənɑ	menep
	this.ACC	see-VADV	they-GEN	one.of.them	very	fear-VADV	tree	top-POSS.3s-DAT	climb.up-VADV
	qɑɕqɑn								
	flee-PST.3s								

On seeing this, one of them became very frightened, and fled by climbing up to the top of a tree.

(5)	ikenɕese	ʉlgæn	buləp	tiz	genæ	suzələp	jɑtqɑn
	second-POSS.3s	die-VADJ	pretend-VADV	quickly	just	be.stretched.out-VADV	lie-PST.3s

The other quickly lay down, and pretended to be dead.

(6) ɑju bu juldɑ jɑtqɑn keʃeneŋ jɑnəndɑ æjlænæ æjlænæ
bear this road-LOC lie-VADJ person-GEN side-POSS.3S-LOC go.around-VADV go.around-VADV
bɑʃlɑræn ɑjɑqlɑrən isnægæn dæ dij ber dæ timijɕæ
head-PL-POSS.3S-ACC leg-PL-POSS.3S-ACC sniff-PST.3s also say-PRS.3s one also touch-NEG-VADV
ʉz julənɑ kitkæn dij
self way-POSS.3S-DAT go-PST.3s say-PRS.3s

The bear circled around the man lying on the road, sniffed at his head and legs, and went on its way without touching him.

(7) ɑju kitkænːæn soŋ ælege keʃe ɑʁɑɕ bɑʃənːɑn tø̞ʃep
bear go-VADJ-ABL after.PP the.aforementioned person tree top-POSS.3S-ABL climb.down-VADV
iptæʃenːæn sorɑʁɑn
friend-POSS.3S-ABL ask-PST.3s

After the bear had left, the first man climbed down from the top of the tree and asked his friend:

(8) ɑju sineŋ qolɑʁənɑ ʃəpərt qənɑ nærsæ æjtːe digæn
bear you.GEN ear-POSS.3S-DAT quietly what say-PST.3s say-PST.3s

‘What did the bear whisper in your ear?’

(9) bu keʃe æjtkæn ɑju miŋɑ monːɑn soŋ ʃuʃəndəj qurqɑq kʃe belæn julʁɑ
this person say-PST.3s bear I.DAT this.ABL after.PP such coward person with.PP road-DAT
ɕəqmɑ dip æjtːe digæn
go.out-NEG.IMP.2s say-VADV say-PST.3s say-PST.3s

The (other) man said: ‘The bear told me not to go travelling with such a coward anymore.’

Abbreviations

1p	1st person plural
1s	1st person singular
2p	2nd person plural
2s	2nd person singular
3p	3rd person plural
3s	3rd person singular
ABL	ablative
ACC	accusative
ADJZ	adjectiviser
AUX	auxiliary
COMP	comparative
COP	copula
DAT	dative
DER	derivation
EQ	equative
GEN	genitive
IMP	imperative
LOC	locative
NMLZ	nominalizer
NEG	negative
OPT	optative
PL	plural
PRIV	privative
POSS	possessive
PP	postposition
PRS	present
PST	past
REL	relationaliser
SIM	similative
VADJ	verbal adjective
VADV	verbal adverb
VOL	voluntative
VN	verbal noun

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Supplementary material

Final statement about the accessibility of the supplementary files — to be finalised after review.

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